Sensitivity Analysis in Population Shares*

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Empirical work often relies on assumptions on the distribution of random variables, such as potential outcomes or latent variables, to identify some parameter of interest. We introduce a unified framework for assessing the sensitivity of a target estimand to relaxing these distributional assumptions for a given share of the population. We derive bounds on the values the estimand can take as this share varies between 0 and 1, and compute the bounds using tractable finite-dimensional programs. We propose plug-in estimators for the bounds and characterize their large sample properties. We also show that our bounds recover exact nonparametric sharp bounds at no additional computational cost, and introduce summary measures that separate misspecification uncertainty from sampling uncertainty and rank studies by their robustness. We apply our framework to assess the robustness of (*i*) average treatment effects in randomized experiments with attrition, (*ii*) counterfactual predictions in structural models, and (*iii*) formula instruments. We provide `R` and `Python` packages implementing all our methods.

Keywords: sensitivity analysis, robustness, partial identification, convex duality.

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1 Introduction

A long tradition has stressed the need for sensitivity analyses to assess the fragility of a study's conclusions (Leamer, 1983; Rosenbaum and Rubin, 1983; Leamer, 1985; Tamer, 2015). Consider a policymaker deciding which of several structural models or natural experiments should inform policy, a manager allocating a finite budget across pilot studies, or a drug regulator weighing the evidence from competing trials. These alternative empirical analyses report point estimates and confidence intervals for a common target parameter, but all rest on untestable, sometimes not even falsifiable, distributional assumptions: that attrition in a randomized experiment is random, that unobserved preference shocks in an indirect utility follow an extreme value distribution, or that the exogenous shocks in a formula instrument are uniformly distributed. When the studies disagree, the decision maker needs to know how much of the disagreement reflects misspecification rather than sampling uncertainty, and two questions arise. Can we quantify how sensitive each conclusion is to distributional assumptions, so that studies can be ranked? And can we do so in units interpretable enough to communicate the ranking to a broad audience of stakeholders?

This paper answers both questions with a framework that assesses the sensitivity of a target estimand to violations of distributional assumptions and measures the extent of a violation by the share of the population that does not satisfy such assumptions. The framework gauges, for instance, how much an average treatment effect would change if nonresponse were nonrandom for 60% of the experimental treated units, how much a counterfactual welfare prediction from a structural model would change if 30% of the agents drew shocks that were not extreme value distributed, or how much an estimated elasticity would change if 40% of the time the exogenous shocks were assigned by a mechanism other than the one the researcher postulated.

Formally, the object of sensitivity analysis is an expectation $\mathbb{E}_{P_*}[g(Z;\theta)]$, where g is a known function, Z is a random vector, θ is a structural parameter disciplined by moment equalities or inequalities, and P_* is the *baseline distribution*, the model the econometrician would use absent any concern for misspecification. In this scenario, our goal is to obtain the smallest and largest values the estimand can take over a nonparametric neighborhood of candidate distributions that allow up to a fraction δ of the population to violate the distributional assumption imposed by P_*. We refer to these two values as the *sensitivity bounds* at δ, to the neighborhood as the *robustness set*, and to δ as the *sensitivity parameter*. In a randomized experiment with attrition, δ bounds the share of treated respondents for whom attrition is not random. In a discrete choice model, it bounds the share of agents whose taste shocks do not follow the postulated distribution. With formula instruments, it bounds the probability that the shocks were assigned by a different mechanism.

We show that the sensitivity bounds are the optimal values of two symmetric infinite-dimensional programs, which in general admit no closed form solution. Using tools from convex analysis (Rockafellar, 1970; Rockafellar and Wets, 2009; Kuhn, Shafiee and Wiesemann, 2025), we recast them as finite-dimensional convex programs, so that the entire path of bounds can be traced as δ ranges between 0 and 1. At $\delta = 0$, the bounds reduce to the value of the estimand under the baseline specification P_*. At $\delta = 1$, we prove that they coincide with the sharp nonparametric bounds,

namely the tightest bounds on the estimand implied by the model's remaining restrictions once the distributional assumption is dropped entirely. As a consequence, our procedure can also be used to compute nonparametric sharp bounds directly from a tractable program whose dimensionality grows only with the number of moment conditions imposed. Put differently, δ trades off informativeness against credibility, in the spirit of *the law of decreasing credibility* of Manski (2003). At $\delta = 0$, the estimand is point identified, but only under the full distributional assumption. At $\delta = 1$, the bounds rest on the model's remaining restrictions alone, but they are at their widest.

Regarding estimation and inference, we propose simple plug-in estimators for the bounds. We allow for both a known baseline, as when P_* is the distribution of taste shocks in a discrete choice model, and an estimated one, as when P_* is the outcome distribution of the treated respondents in a randomized experiment. We establish the large sample properties of the estimators by showing their consistency, deriving their limiting distribution, and proposing a bootstrap procedure to construct valid confidence intervals for the bounds.

We also propose summary measures that separate misspecification uncertainty from sampling uncertainty and make studies comparable. Together with the bounds, they answer the two questions raised in our opening paragraphs. Finally, we apply the framework to the online tutoring experiment of Carlana and La Ferrara (2025), in which attrition threatens identification, to the marital college premium application of Christensen and Connault (2023), and to the formula instruments of Borusyak and Hull (2023), who estimate the employment effects of market access growth from Chinese high-speed rail construction.

Related literature. Some recent work studies the sensitivity of counterfactuals in structural models. In Christensen and Connault (2023), the values a counterfactual takes are bounded as the distribution of latent variables ranges over a ϕ-divergence neighborhood of a parametric specification while the structural model's moment restrictions are maintained. Gu and Russell (2024) bound scalar counterfactuals in partially identified structural models over Wasserstein neighborhoods of the latent distribution using random set theory, and Lieber, Tebaldi and Torgovitsky (2026) bound counterfactuals in nonparametric discrete choice models when the latent density lies in a pointwise band around a known reference density, using tools from graph theory and mechanism design. This paper differs from these works mainly in that our sensitivity parameter δ admits a population share interpretation and our framework applies beyond the structural models these papers consider.

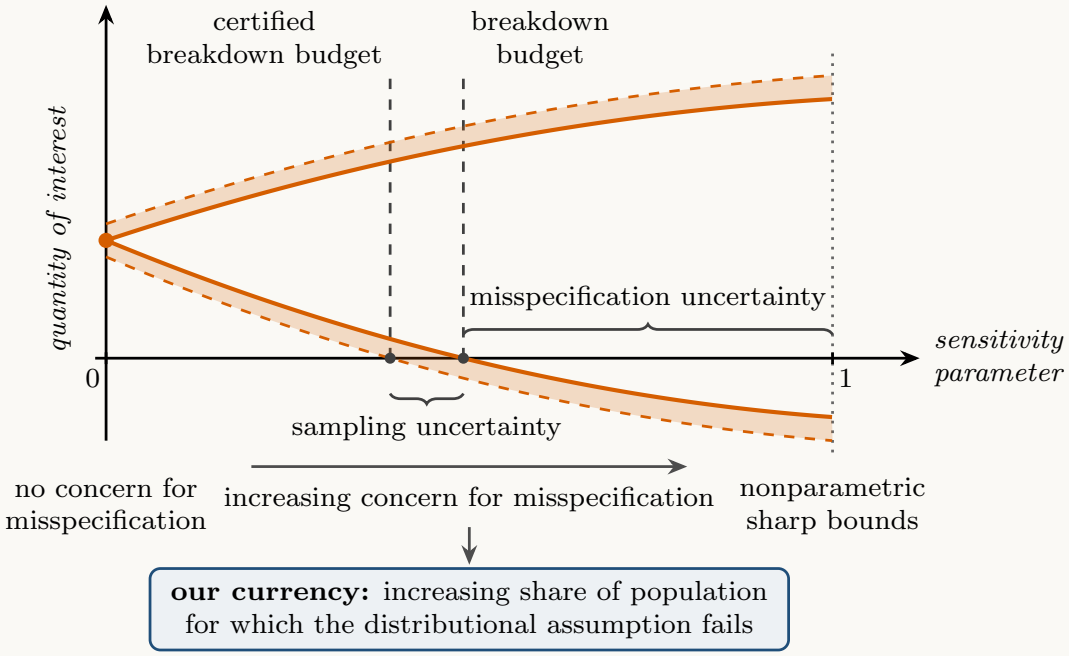
Our framework is also related to a large body of work that develops sensitivity exercises within specific identification problems: contaminated and corrupted sampling (Horowitz and Manski, 1995); missing data and attrition (Kline and Santos, 2013; Ober-Reynolds, 2026); selection on unobservables and unmeasured confounding (Rosenbaum and Rubin, 1983; Imbens, 2003; Altonji, Elder and Taber, 2005; Masten and Poirier, 2018; Bonvini and Kennedy, 2022; Masten, Poirier and Zhang, 2024; Masten, Poirier and Ren, 2025); the exclusion and exogeneity of instruments (Masten and Poirier, 2021; Diegert, Masten and Poirier, 2026); the distribution of the exogenous shocks that generate formula instruments (Sheng and Torgovitsky, 2026); random assignment and rank invariance (Masten and Poirier, 2020); and parallel trends in difference-in-differences designs (Rambachan and Roth, 2023). Relative to these papers, we provide a single framework that nests

many of the exercises described above, can include moment restrictions and structural parameters, provides a general estimation and inference theory, and allows the sensitivity parameter to maintain an analogous population share interpretation for a wide family of settings. In this sense, our sensitivity parameter is a *common currency* for misspecification.

Another strand of literature focused on local sensitivity analysis, which measures how quantities of interest respond to infinitesimal perturbations of the model or of its moment conditions (Kitamura, Otsu and Evdokimov, 2013; Andrews, Gentzkow and Shapiro, 2017; Mukhin, 2018; Andrews, Gentzkow and Shapiro, 2020; Armstrong and Kolesár, 2021; Bonhomme and Weidner, 2022), whereas we allow for global misspecification, in that δ can be large and does not shrink as the sample size grows. We show how the two approaches can be linked via the shadow price of robustness, the derivative of the sensitivity bounds with respect to δ, which is a local sensitivity measure at small budgets.

Finally, on the methodological side, we contribute to a line of work in economics that uses convex duality to turn optimization problems over infinite-dimensional sets of distributions into tractable ones (Hansen and Sargent, 2001; Ekeland et al., 2010; Schennach, 2014). Closest to our work, Christensen and Connault (2023) use duality to recast the optimization of a counterfactual over a ϕ-divergence neighborhood, subject to moment restrictions, as a finite-dimensional convex program. Their duality results are stated for divergences generated by strictly convex and differentiable functions, a class that excludes the distance on which our neighborhoods are built.

Figure 1: *Sensitivity analysis in population shares: a visual roadmap.*



Paper organization. Figure 1 collects the main output of a prototypical sensitivity exercise conducted with our framework and organizes the rest of the paper. Section 2 presents three leading applications that motivate our sensitivity exercise. Section 3 formally defines the framework nesting these applications and the optimization problems behind the thick lines, and explains why moving

rightward on the horizontal axis raises the share of the population for which the distributional assumption imposed by P_* is allowed to fail. Section 4 shows how to compute, estimate, and conduct inference on the sensitivity bounds, and thereby delivers the thick lines and the dashed bands around them. Section 5 proves that at $\delta = 1$ the bounds coincide with the exact nonparametric sharp bounds and discusses how to use them in practice by means of summary measures, such as the breakdown and certified breakdown budgets of Figure 1. The breakdown budget is the smallest share of the population for which the sensitivity bounds contain zero, whereas the certified breakdown budget, which also accounts for sampling uncertainty, is the smallest share for which the dashed bands do. Section 6 deploys the framework in the three empirical applications introduced in Section 2, and Section 7 concludes. The supplementary materials collect all proofs, establish the existence of solutions to the dual and outer problems, relax the simplifying conditions of Section 4, allow for moment inequalities and nuisance parameters, and work out further applications.

Notation. For a set $\mathcal{A}$, $\Delta(\mathcal{A})$ denotes the set of probability distributions on $\mathcal{A}$ and, for $P, Q \in \Delta(\mathcal{A})$, $\mathsf{TV}(P \,\|\, Q) := \sup_{A \subseteq \mathcal{A}} |P(A) - Q(A)| \in [0,1]$ denotes the total variation distance between them. We write $P \ll Q$ when P is absolutely continuous with respect to Q, $\mathrm{d}P/\mathrm{d}Q$ for the corresponding density, $\mathrm{supp}(P)$ for the support of P, $Pf := \mathbb{E}_P[f(Z)]$ for a random vector Z distributed according to P and a P-integrable function f, and $P\text{-}\operatorname{ess\,sup}_z f(z)$ for the smallest $c \in [-\infty, +\infty]$ such that $f \leq c$ holds P-almost surely. We let $[n] := \{1, \ldots, n\}$ for $n \in \mathbb{N}$, $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$, $\mathbb{R}_+ := [0, \infty)$, 0_d be the d-dimensional zero vector, $\mathbb{1}\{\cdot\}$ the indicator function, $x^\top$ the transpose of x, and $\|\cdot\|$ the Euclidean norm. For a convex set $C \subseteq \mathbb{R}^d$, $\mathrm{int}(C)$ and $\mathrm{ri}(C)$ denote its interior and relative interior. Finally, $\rightsquigarrow$ denotes weak convergence, $o_{\mathbb{P}}(1)$ a term converging to zero in probability, and $\mathsf{N}(0, \Sigma)$ the Gaussian distribution with covariance matrix Σ.

2 Motivating Examples

In this section, we illustrate how three common economic applications rest on untestable distributional assumptions and are thus amenable to sensitivity analysis: a randomized controlled trial (RCT, henceforth) with attrition, counterfactuals in structural models following Christensen and Connault (2023), and the recentered instrumental variables estimator of Borusyak and Hull (2023). Many economically relevant scenarios beyond these three fit the framework (see Section SA6 of the supplementary materials).

Example 2.1 (RCTs with attrition)**.** Consider a randomized experiment that grants half of the applicants access to a free job training course and then, two years later, runs a follow-up survey to record each applicant's weekly wage.[1] Wages, however, are observed only for the applicants who follow up. Worse, trainees stay in touch with the agency while the control-group applicants are the hardest to track down, so respondents may end up being selected on particular dimensions, confounding the effects of the program. Practitioners typically assume attrition is ignorable or,

[1] A famous program of this kind is Job Corps, which offers vocational training to disadvantaged youth aged 16–24 and remains active today with more than 120 campuses across the United States and Puerto Rico, roughly 25,000 students enrolled at a time, and a budget of about $1.8 billion for fiscal year 2026 (U.S. Congress, 2026).

equivalently, that nonresponse is as good as random, an assumption known as "missing completely at random" (MCAR, henceforth) that we define precisely below.[2]

Formally, consider a population described by $(Y(0), Y(1), S(0), S(1), D) \sim P_0$, where $D \in \{0,1\}$ is a binary treatment, $Y(d) \in \mathcal{Y}$, $d \in \{0,1\}$, are potential outcomes taking values in a closed set $\mathcal{Y} \subseteq \mathbb{R}$, $S(d) \in \{0,1\}$, $d \in \{0,1\}$, are potential response indicators, and P_0 satisfies random assignment of D, that is $D \perp\!\!\!\perp (S(0), S(1), Y(0), Y(1))$. Observed and potential quantities are linked by $Y = DY(1) + (1-D)Y(0)$ and $S = DS(1) + (1-D)S(0)$. The econometrician observes (YS, S, D) for every applicant, as the outcome Y is observed only for respondents, that is, when $S = 1$. The pair of potential response indicators partitions the population into four groups:

$$G := \begin{cases} \mathsf{AO} & \text{if } (S(0), S(1)) = (1,1), \\ \mathsf{C} & \text{if } (S(0), S(1)) = (0,1), \\ \mathsf{NO} & \text{if } (S(0), S(1)) = (0,0), \\ \mathsf{DF} & \text{if } (S(0), S(1)) = (1,0), \end{cases}$$

namely the *always-observed*, who respond regardless of the arm they are assigned to; the *compliers*, who respond only if treated; the *never-observed*, who drop out of the sample regardless of the arm; and the *defiers*, who respond only if untreated.[3] As a parameter of interest, we consider the average treatment effect (ATE, henceforth) for the always-observed,[4] which is defined as

$$\tau_0 := \mathbb{E}_{P_0}[Y(1) - Y(0) \mid G = \mathsf{AO}].$$

For simplicity, we assume throughout that treatment weakly encourages response, i.e., $S(1) \geq S(0)$ almost surely (Mono), an assumption known as "monotonicity," which in this notation reads $P_0[G = \mathsf{DF}] = 0$. In Remark SA-15 of the supplementary materials we explain how to conduct the same sensitivity exercise without Mono. Absent defiers, the respondents in the control arm are exactly the always-observed, hence

$$\mu^{\mathsf{AO}}(0) := \mathbb{E}_{P_0}[Y(0) \mid G = \mathsf{AO}] = \mathbb{E}_{P_0}[Y \mid D = 0, S = 1] \tag{2.1}$$

is nonparametrically identified, and we can focus on the treated arm.

[2]Perhaps surprisingly, this practice is the norm rather than the exception. In a systematic review of survey-based articles published in the top-five economics journals, Dutz, Huitfeldt, Lacouture, Mogstad, Torgovitsky and van Dijk (2026) find that nonresponse rates average 50% when researchers collect their own data and 19% when using pre-existing household surveys. Despite that, nearly half of the articles do not discuss nonresponse bias at all, while the remaining articles either assert that nonresponse is ignorable or correct for it by reweighting on observables. Linking their own survey to administrative records on the full population, they document substantial nonresponse bias that survives adjustment for observables, even in a survey run by a national statistical agency. There is further empirical evidence in economics that agents, when given the choice of whether to report, make that decision on the basis of observable or unobservable characteristics (Mu, Rubin and Zou, 2026; Bollinger, Hirsch, Hokayem and Ziliak, 2019; Crippa, Berg and Rigobon, 2025).

[3]This taxonomy deliberately mirrors the imperfect compliance taxonomy of Angrist, Imbens and Rubin (1996), with response to the follow-up survey playing the role of treatment take-up.

[4]Many more causal parameters are identified under MCAR. We focus on this particular one because it is often the object of interest of the attrition literature and because it can be nonparametrically partially identified without bounded $\mathcal{Y}$ when MCAR does not hold. We show how to conduct sensitivity analysis when the ATE over the whole population is the target estimand in Section SA6.4.3 of the supplementary materials.

In the treated arm, the data identify the conditional distribution of Y given $D = 1$ and $S = 1$ under P_0, which we denote by P_{T} and whose mean we denote by $\mu_{\mathsf{T}} := \mathbb{E}_{P_{\mathsf{T}}}[Y]$, but not who among the treated respondents is a complier and who an always-observed. Let P_{C} and P_{AO} denote the conditional distributions of $Y(1)$ given $G = \mathsf{C}$ and given $G = \mathsf{AO}$ under P_0, respectively, and let

$$\pi := P_0[G = \mathsf{C} \mid D = 1, S = 1] = \frac{P_0[S = 1 \mid D = 1] - P_0[S = 1 \mid D = 0]}{P_0[S = 1 \mid D = 1]} \tag{2.2}$$

be the complier share among treated respondents, where the second equality follows from random assignment and Mono, so that π is identified by comparing the response rates of the two arms. We assume that $P_0[S = 1 \mid D = 0] > 0$, that treatment strictly raises the response rate, and that $\mathbb{E}_{P_0}[|Y(d)|] < \infty$, $d \in \{0, 1\}$, so that $\pi \in (0, 1)$ and the means above are finite. Since $S(1) = 1$ if and only if $G \in \{\mathsf{AO}, \mathsf{C}\}$, random assignment yields the mixture decomposition

$$P_{\mathsf{T}} = \pi P_{\mathsf{C}} + (1 - \pi) P_{\mathsf{AO}}, \tag{2.3}$$

in which neither component is identified. The decomposition (2.3) is key for two reasons. First, MCAR, that is, the assumption that response is independent of potential outcomes, $G \perp\!\!\!\perp (Y(0), Y(1))$, implies that the distribution of $Y(1)$ is the same for compliers and always-observed, so that in the treated arm MCAR amounts to

$$P_{\mathsf{C}} = P_{\mathsf{AO}} = P_{\mathsf{T}}, \tag{2.4}$$

which identifies the unobserved complier distribution with the observed distribution of treated respondents. Second, (2.3) ties the always-observed to the compliers: for any candidate complier distribution $Q \in \Delta(\mathcal{Y})$, the mixture identity forces the always-observed distribution to be $(P_{\mathsf{T}} - \pi Q)/(1 - \pi)$, whose mean is $(\mu_{\mathsf{T}} - \pi\, \mathbb{E}_Q[Y])/(1 - \pi)$. Subtracting (2.1), we obtain the map

$$\tau(Q) := \frac{\mu_{\mathsf{T}} - \pi\ \mathbb{E}_Q[Y]}{1 - \pi} - \mu^{\mathsf{AO}}(0), \qquad Q \in \Delta(\mathcal{Y}), \tag{2.5}$$

in which $\mathbb{E}_Q[Y]$ is the only quantity not identified from the data. At the true complier distribution the map recovers the parameter of interest, $\tau(P_{\mathsf{C}}) = \tau_0$. Under MCAR, (2.4) gives $\tau_0 = \tau(P_{\mathsf{T}})$, so that τ_0 is point identified by the difference in mean outcomes among respondents (Rubin, 1976),

$$\tau_{\mathsf{MCAR}} := \tau(P_{\mathsf{T}}) = \mathbb{E}_{P_0}[Y \mid D = 1, S = 1] - \mathbb{E}_{P_0}[Y \mid D = 0, S = 1].$$

However, MCAR is untestable and hard to justify by economic theory. Hence, a stakeholder might be interested in quantifying how far τ_0 can lie from τ_{MCAR} when MCAR fails.

The way we model relaxations of MCAR is by requiring that the outcome distributions of compliers and always-observed be close rather than equal. Put differently, we consider candidate pairs $(Q_{\mathsf{C}}, Q_{\mathsf{AO}})$ of distributions on $\mathcal{Y}$ that satisfy $P_{\mathsf{T}} = \pi Q_{\mathsf{C}} + (1 - \pi) Q_{\mathsf{AO}}$ and $\mathsf{TV}(Q_{\mathsf{C}} \,\|\, Q_{\mathsf{AO}}) \le \delta$. As we show in Section 3, with this formulation, δ is the share of treated respondents or, equivalently, the share of compliers who may not satisfy MCAR.

The pair $(Q_{\mathsf{C}}, Q_{\mathsf{AO}})$ is fully described by its first component. Indeed, the mixture constraint (3.3) gives that $Q_{\mathsf{AO}} = (P_{\mathsf{T}} - \pi Q_{\mathsf{C}})/(1 - \pi)$ and that $\mathsf{TV}(Q_{\mathsf{C}} \,\|\, P_{\mathsf{T}}) \le (1 - \pi)\delta$.[5] This reformulation

[5]The equivalence follows from the identity $\mathsf{TV}(Q_{\mathsf{C}} \,\|\, P_{\mathsf{T}}) = (1 - \pi)\mathsf{TV}(Q_{\mathsf{C}} \,\|\, Q_{\mathsf{AO}})$, which total variation inherits from being a norm-induced metric and which ϕ-divergences do not enjoy in general.

replaces the unidentified reference distribution Q_{AO} by the identified P_{T}, and lets us define the set of candidate complier distributions as

$$\mathcal{P}_{\mathsf{RCT}}(\delta) := \big\{Q \in \Delta(\mathcal{Y}) : \mathsf{TV}(Q \,\|\, P_{\mathsf{T}}) \leq (1-\pi)\delta, \\ P_{\mathsf{T}} = \pi Q + (1-\pi)Q_{\mathsf{AO}} \text{ for some } Q_{\mathsf{AO}}\big\}. \tag{2.6}$$

The sensitivity bounds are then the extreme values that the causal estimand in (2.5) can take over the robustness set defined in (2.6). Formally, for $\delta \in [0,1]$

$$\underline{\tau}(\delta) := \inf_{Q \in \mathcal{P}_{\mathsf{RCT}}(\delta)} \tau(Q) \qquad \text{and} \qquad \overline{\tau}(\delta) := \sup_{Q \in \mathcal{P}_{\mathsf{RCT}}(\delta)} \tau(Q).$$

♦

Example 2.2 (Counterfactuals in structural models)**.** Counterfactual predictions from structural models depend on the distribution specified for latent variables such as taste shocks, unobserved heterogeneity, or productivity shocks, which is typically not nonparametrically identifiable and is therefore chosen for tractability. A stakeholder may want to know how sensitive the predictions are to this untestable choice. As a running example, we revisit one of the leading empirical applications of Christensen and Connault (2023) (CC23, henceforth): the marital college premium in the marriage-matching model that Chiappori, Salanié and Weiss (2017) estimate on American Community Survey data.

Consider a frictionless one-to-one matching model with transferable utility. Men and women each belong to one of $J \in \mathbb{N}$ education groups, indexed by $[J] := \{1, \ldots, J\}$, which we term "types." We let $j = 0$ denote the outside option of remaining unmatched. A type-$b \in [J]$ woman receives utility ν_{0b} if unmatched and $t_{jb} + \nu_{jb}$ if matched with a type-$j \in [J]$ man, where t_{jb} is the systematic component of match utility and ν_{jb} captures idiosyncratic marital preferences. For simplicity, we assume that the distribution of females' preference shocks does not depend on their type, so we drop the b subscript and consider a single random vector $U = (\nu_0, \ldots, \nu_J)^\top$.[6] We let $\widetilde{P}$ denote the distribution of latent preference shocks $(\nu_0, \ldots, \nu_J)$ postulated by the econometrician, which we assume to be a Gumbel (type I extreme value) distribution.[7]

We fix two female types $b, b' \in [J]$, whose marital college premium is the object of interest, and let $\varrho \in \mathbb{R}^{2J}$ collect their observed marriage probabilities with each male type. The model pins down the structural parameter $\vartheta := (t_{jb}, t_{jb'}, j \in [J])$, ranging over a parameter space $\Theta \subseteq \mathbb{R}^{2J}$, via a set of moment equalities

$$\mathbb{E}_{\widetilde{P}}[s(U;\vartheta)] = (\varrho, 0_{2(J+1)}), \qquad s(U;\vartheta) = (s_1(U;\vartheta)^\top, s_2(U;\vartheta)^\top)^\top,$$

where $s(U;\vartheta)$ is the moment function, which we now describe in detail. First,

$$s_1(U;\vartheta) = \begin{bmatrix} \left(\mathbb{1}\left\{j = \underset{j' \in \{0,\ldots,J\}}{\arg\max} \left(t_{j'b} + \nu_{j'}\right)\right\}\right)_{j=1}^{J} \\ \left(\mathbb{1}\left\{j = \underset{j' \in \{0,\ldots,J\}}{\arg\max} \left(t_{j'b'} + \nu_{j'}\right)\right\}\right)_{j=1}^{J} \end{bmatrix}.$$

[6] See Section SA6.2, and Remark SA-16 in particular, in the supplementary materials for a discussion of this common-shock representation and its alternatives.

[7] Another common choice is to assume that $\widetilde{P}$ is a standard Gaussian distribution, which gives a probit instead of a logit discrete choice model.

Ties are broken toward the lowest indexed alternative. Therefore, the moment conditions $\mathbb{E}_{\widetilde{P}}[s_1(U;\vartheta)] = \varrho$ equate the observed marriage probabilities ϱ for female types b and b' to the model implied choice probabilities under $\widetilde{P}$. Note that because the moment function must be a known function of (U,ϑ) alone, the observed marriage probabilities ϱ enter as the target of the moment conditions rather than as an argument of s_1.

Second, one may impose some normalizations on the female shocks through

$$s_2(U;\vartheta) = (\nu_j, \nu_j^2 - \pi^2/6)_{j=0}^{J}.$$

Therefore, the moment conditions $\mathbb{E}_{\widetilde{P}}[s_2(U;\vartheta)] = 0_{2(J+1)}$ impose that each shock must have mean zero and variance $\pi^2/6$ under $\widetilde{P}$, the same variance as a standard Gumbel distribution.

Consider the marital college premium (i.e., the difference in indirect utilities) for female types b and b',

$$k(\vartheta;\widetilde{P}) := \mathbb{E}_{\widetilde{P}}[f(U;\vartheta)], \quad f(U;\vartheta) = \max_{j\in\{0,\ldots,J\}}\{t_{jb} + \nu_j\} - \max_{j\in\{0,\ldots,J\}}\{t_{jb'} + \nu_j\},$$

where $t_{0b} = t_{0b'} = 0$. A stakeholder interested in the output of the model above may ask what happens to the premium if not all women draw their shocks from a Gumbel distribution. To answer this question, we let P range over the robustness set

$$\begin{aligned}\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta) := \big\{P \in \Delta(\mathbb{R}^{J+1}) : \mathbb{E}_P[s(U;\vartheta)] = (\varrho, 0_{2(J+1)}),\\ P = (1-\delta)\widetilde{P} + \delta R,\ R \in \Delta(\mathbb{R}^{J+1})\big\},\end{aligned} \tag{2.7}$$

which collects all the populations of women that (i) reproduce the observed marriage probabilities and satisfy the imposed normalizations and (ii) consist of a share $\delta \in [0,1]$ of women with arbitrary shocks and a share $1-\delta$ of women with Gumbel shocks. The sensitivity bounds on the premium are then

$$\underline{\mathsf{k}}(\delta) := \inf_{\vartheta\in\Theta} \inf_{P\in\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta)} \mathbb{E}_P[f(U;\vartheta)], \qquad \overline{\mathsf{k}}(\delta) := \sup_{\vartheta\in\Theta} \sup_{P\in\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta)} \mathbb{E}_P[f(U;\vartheta)],$$

where the inner infimum and supremum are set to $+\infty$ and $-\infty$, respectively, whenever $\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta)$ is empty. Therefore, the values of ϑ at which no distribution within δ of the Gumbel baseline matches the observed marriage probabilities are infeasible. ♦

Example 2.3 (Recentered IV)**.** Many causal questions in economics involve treatments or instruments that combine multiple sources of variation according to a known formula. Examples include shift-share (or Bartik) instruments, treatments capturing spillovers in social or transportation networks, or simulated instruments for policy eligibility (see Borusyak, Hull and Jaravel, 2025, for a review). Borusyak and Hull (2023) (BH23, henceforth) point out that exogeneity of the shocks entering the formula does not, by itself, deliver a consistent instrumental variables estimator, because units are nonrandomly exposed to the shocks. The solution BH23 propose is to recenter the instrument, subtracting from it the *expected instrument*, hat is, the average instrument across counterfactual shocks. Computing the expected instrument requires the researcher to specify the

distribution of the shocks, or "shock assignment process." In a randomized experiment the assignment process is given by the randomization protocol, but in natural experiments the assignment process is rarely known or justified by economic theory. A stakeholder may accordingly ask how much the recentered IV estimate would change under a different assignment process.

Formally, let $i \in [n], n \in \mathbb{N}$, index the units of the sample and consider the causal or structural model

$$y_i = \beta x_i + \varepsilon_i,$$

where $y_i \in \mathbb{R}$ is a demeaned outcome, $x_i \in \mathbb{R}$ is a demeaned endogenous regressor, $\varepsilon_i \in \mathbb{R}$ is an unobserved residual, and $\beta \in \mathbb{R}$ is the parameter of interest. Let $v = (v_1, \ldots, v_K) \in \mathcal{V} \subseteq \mathbb{R}^K$, $K \in \mathbb{N}$, be a vector of exogenous shocks, let w collect predetermined covariates of unrestricted dimension, and let

$$z_i := f_i(v; w)$$

be the candidate instrument of unit $i \in [n]$, where $f_i(\cdot\,; w) : \mathcal{V} \to \mathbb{R}$ are known, bounded, non-stochastic functions.[8] BH23 assume that the shocks are exogenous, that is,

$$v \perp\!\!\!\perp (\varepsilon_1, \ldots, \varepsilon_n) \mid w, \tag{2.8}$$

and show that this alone does not make z_i orthogonal to ε_i, because units are nonrandomly exposed to the shocks through the formula.

Letting $\widetilde{P} \in \Delta(\mathcal{V})$ denote the assignment process postulated by the econometrician, the expected instruments are $\mu_i(\widetilde{P}) := \mathbb{E}_{\widetilde{P}}[f_i(v; w)]$ and the recentered instruments are $\widetilde{z}_i := z_i - \mu_i(\widetilde{P})$, $i \in [n]$.[9] The recentered IV estimator of BH23 is then

$$\widehat{\beta}_* := \frac{\sum_{i=1}^n \widetilde{z}_i\, y_i}{\sum_{i=1}^n \widetilde{z}_i\, x_i}. \tag{2.9}$$

The setup also covers the case in which the treatment itself is generated by the formula, so that $z_i = x_i$ as in the empirical application of BH23 that we revisit in Section 6.3.

Our sensitivity exercise maintains exogeneity of the shocks in (2.8) and relaxes the assumption that their assignment process is known and equal to $\widetilde{P}$. For this exercise, we hold the realized sample (outcomes, treatments, covariates, and the realized shocks $\check{v}$) fixed, so that the recentered estimate is a function of the assignment process alone. Because (2.9) depends on $\widetilde{P}$ only through the expected instruments, replacing the postulated assignment process by a candidate one $P \in \Delta(\mathcal{V})$ produces the estimate

$$\widehat{\beta}(P) := \frac{G_y(P)}{G_x(P)}, \tag{2.10}$$

[8] Boundedness holds in the empirical application of Section 6.3, where $\mathcal{V}$ is a finite set.

[9] We assume for simplicity that the distribution of $v \mid w$ does not depend on w, as is the case for the application we consider in Section 6.3.

where

$$G_y(P) := g_y(\check{v}) - \mathbb{E}_P[g_y(v)], \qquad g_y(\cdot) := \sum_{i=1}^{n} y_i f_i(\cdot\,; w),$$
$$G_x(P) := g_x(\check{v}) - \mathbb{E}_P[g_x(v)], \qquad g_x(\cdot) := \sum_{i=1}^{n} x_i f_i(\cdot\,; w).$$

By construction $\widehat{\beta}(\widetilde{P}) = \widehat{\beta}_*$, so the exercise is centered at the reported estimate. For any $\delta \in [0,1]$, we let the candidate assignment process P range over

$$\mathcal{P}_{\mathsf{FI}}(\delta) := \big\{P \in \Delta(\mathcal{V}) :\ P = (1-\delta)\widetilde{P} + \delta R \ \text{ for some } R \in \Delta(\mathcal{V})\big\}. \tag{2.11}$$

Thus, up to a share δ of the shock draws may be assigned by a mechanism other than the one the researcher postulated. Throughout the exercise we maintain that the recentered first stage is uniformly bounded away from zero over the largest robustness set, that is,

$$\inf_{P \in \mathcal{P}_{\mathsf{FI}}(1)} |G_x(P)| > 0. \tag{2.12}$$

As P ranges over $\mathcal{P}_{\mathsf{FI}}(1) = \Delta(\mathcal{V})$, the expectation $\mathbb{E}_P[g_x(v)]$ ranges over an interval with endpoints $\inf_{v' \in \mathcal{V}} g_x(v')$ and $\sup_{v' \in \mathcal{V}} g_x(v')$, so that condition (2.12) holds if and only if $g_x(\check{v})$ lies strictly above this supremum or strictly below this infimum. In particular, it requires $\check{v} \notin \mathcal{V}$. Because the robustness sets are nested, condition (2.12) guarantees that the estimator is well defined under every candidate assignment process and at every budget.[10] Since the analysis is conditional on the realized sample, the left-hand side of (2.12) is directly computable, and in the empirical application of Section 6.3 we verify that it holds. Finally, we define the sensitivity bounds as

$$\underline{\beta}(\delta) := \inf_{P \in \mathcal{P}_{\mathsf{FI}}(\delta)} \widehat{\beta}(P), \qquad \overline{\beta}(\delta) := \sup_{P \in \mathcal{P}_{\mathsf{FI}}(\delta)} \widehat{\beta}(P).$$

At $\delta = 0$ the robustness set reduces to $\{\widetilde{P}\}$ and both bounds collapse to $\widehat{\beta}_*$. ♦

3 Setup

This section formally defines our framework for sensitivity analysis in terms of population shares not satisfying the distributional assumption, and shows how the three leading applications of Section 2 fit into it despite their apparent differences: the first has no structural parameter but an estimated baseline, the second has structural parameters but a known baseline distribution, and the third has a known baseline and no structural parameter. In the supplementary materials we present our fully fledged framework, which also allows for moment inequalities and external nuisance parameters.

Estimand. Let $\Theta \subseteq \mathbb{R}^{\mathsf{d}_\theta}, \mathsf{d}_\theta \in \mathbb{N}_0$ be the parameter space, $\mathcal{Z} \subseteq \mathbb{R}^{\mathsf{d}_z}, \mathsf{d}_z \in \mathbb{N}$ a closed set collecting the possible realizations of a random vector Z. The quantity of interest is a scalar expectation over a distribution $P \in \Delta(\mathcal{Z})$ of a known function $g : \mathcal{Z} \times \Theta \to \mathbb{R}$

$$\mathbb{E}_P[g(Z;\theta)] = \int_{\mathcal{Z}} g(z;\theta)\,\mathrm{d}P(z). \tag{3.1}$$

[10] In Section SA6.3 of the supplementary materials we relax this assumption.

Leading examples are a counterfactual in a structural model, where θ plays the role of a structural parameter, a causal quantity such as an average treatment effect, or an estimated elasticity.

Moment restrictions. Let $m : \mathcal{Z} \times \Theta \to \mathbb{R}^{\mathsf{d}_m}$, $\mathsf{d}_m \in \mathbb{N}_0$ be a known moment function and $\vec{\rho}_0 \in \mathbb{R}^{\mathsf{d}_m}$ a vector of target moments. We require that

$$\mathbb{E}_P[m(Z;\theta)] = \vec{\rho}_0. \tag{3.2}$$

In a structural model these are the conditions that identify the structural parameter θ, for instance by matching model implied choice probabilities with observed ones, while in a randomized experiment they can carry moments of pre-treatment covariates added to tighten the bounds. We allow for the components of $\vec{\rho}_0$ to be either known or estimated. Section 4.2 makes the distinction explicit. When no moment function is present, we set $\mathsf{d}_m = 0$ and $\mathbb{R}^0 = \{0\}$, interpret (3.2) as vacuous, and suppress m and $\vec{\rho}_0$ from the notation.

Baseline distribution. The econometrician postulates that Z is distributed according to a *baseline distribution* P_*, the distribution she would use absent any concern for misspecification. We let $P_* \in \mathcal{P}_\theta(\mathcal{Z})$, where $\mathcal{P}_\theta(\mathcal{Z})$ denotes the set of distributions in $\Delta(\mathcal{Z})$ under which $|g(\cdot\,;\theta)|$ and $\|m(\cdot\,;\theta)\|$ are integrable at $\theta \in \Theta$. For brevity, we omit the dependence on θ and write $\mathcal{P}(\mathcal{Z}) \equiv \mathcal{P}_\theta(\mathcal{Z})$. We reserve P_0 for the true distribution of Z. The baseline may be a parametric specification for a distribution that is not nonparametrically identifiable (e.g., a Gumbel distribution for idiosyncratic taste shocks) or an identified feature of P_0 (e.g., the outcome distribution of observed respondents), in which case it must be estimated. In either case the identifying assumption is the equality between the law of Z and P_*, and the sensitivity exercise relaxes it. Both cases are covered by our framework.

Distributional assumptions and robustness sets. Different assumptions are represented by different distributions in $\mathcal{P}(\mathcal{Z})$. To control the degree of relaxation of the distributional assumption embedded in P_*, or equivalently the degree of misspecification, we introduce a *sensitivity parameter* $\delta \in [0,1]$ to bound the distance between the baseline distribution P_* and any candidate alternative distribution $P \in \mathcal{P}(\mathcal{Z})$. In this paper, the distance is measured in terms of the share of the population in the alternative distribution P for which the distributional assumption embedded in P_* may fail. We call the set of distributions that satisfy the moment restrictions in (3.2) and are within a distance δ of the baseline distribution a *robustness set*. We consider two types of robustness sets, each suited to a different setting among those presented in Section 2.

First, consider settings where the statistical model naturally yields a mixture constraint

$$P_* = \kappa P + (1-\kappa)R, \qquad \text{for some } R \in \mathcal{P}(\mathcal{Z}), \tag{3.3}$$

for some known $\kappa \in (0,1)$. In words, the population described by P_* consists of two sub-populations, or groups: a κ-share of its units belongs to the group whose distribution is P and the remaining $(1-\kappa)$-share belongs to a second group whose distribution is R. This is the case of Example 2.1, where (2.3) shows that the observed treated units are a mixture of two groups, the compliers and the always-observed units, with P playing the role of the complier distribution, R that of the

always-observed distribution, and $\kappa = \pi$ the complier share in (2.2). In this case, we define the *total variation robustness set* as

$$\mathcal{P}_{\mathsf{TV}}(\theta;\delta) := \big\{P \in \mathcal{P}(\mathcal{Z}) : \mathsf{TV}(P \,\|\, R) \le \delta,\ \mathbb{E}_P[m(Z;\theta)] = \vec{\rho}_0,\\ P_* = \kappa P + (1-\kappa)R \text{ for some } R \in \mathcal{P}(\mathcal{Z})\big\}. \tag{3.4}$$

The reason (3.4) is appealing is that δ is the share of the population for which the distributional assumption in P_* may fail. The interpretation of δ just given deserves some elaboration. The constraint $\mathsf{TV}(P \,\|\, R) = \delta$ only tells us that the distributions P and R of the two groups share a fraction $1-\delta$ of their probability mass, in the sense that they admit the mixture representations $P = (1-\delta)C + \delta Q_P$ and $R = (1-\delta)C + \delta Q_R$ for a common distribution $C \in \mathcal{P}(\mathcal{Z})$ and some group-specific distributions $Q_P, Q_R \in \mathcal{P}(\mathcal{Z})$. Then, substituting the two representations above into the mixture constraint (3.3), we get

$$P_* = (1-\delta)\, C + \delta\left(\kappa Q_P + (1-\kappa) Q_R\right).$$

Put differently, for a $(1-\delta)$-share of the population, drawn from the common distribution C, the distribution of Z does not depend on the group a unit belongs to, and only the remaining δ-share, drawn from the group-specific $\kappa Q_P + (1-\kappa)Q_R$, is allowed to differ across groups. The distributional assumption embedded in the baseline, namely that the two groups are drawn from the same distribution, is thus recovered at $\delta = 0$ and is gradually relaxed as δ grows.

Second, when the statistical model does not directly yield a mixture constraint, we consider a *contamination robustness set* in the spirit of Huber (1964), which is defined as

$$\mathcal{P}_{\mathsf{C}}(\theta;\delta) := \big\{P \in \mathcal{P}(\mathcal{Z}) : P = (1-\delta)P_* + \delta R \text{ for some } R \in \mathcal{P}(\mathcal{Z}),\\ \mathbb{E}_P[m(Z;\theta)] = \vec{\rho}_0\big\}. \tag{3.5}$$

In words, a distribution P is a valid candidate distribution if it is a mixture of the baseline distribution P_* and an arbitrary distribution R, with δ being the share of the population that is drawn from R. In this case, the interpretation of δ is identical to the one given above for the total variation robustness set.

Sensitivity bounds. The *sensitivity bounds* are the extreme values the estimand (3.1) can take over the robustness sets (3.4) or (3.5), depending on the application:

$$\underline{\mathsf{b}}_{\diamondsuit}(\delta) := \inf_{\theta\in\Theta} \inf_{P\in\mathcal{P}_{\diamondsuit}(\theta;\delta)} \mathbb{E}_P[g(Z;\theta)] \quad \text{and} \quad \overline{\mathsf{b}}_{\diamondsuit}(\delta) := \sup_{\theta\in\Theta} \sup_{P\in\mathcal{P}_{\diamondsuit}(\theta;\delta)} \mathbb{E}_P[g(Z;\theta)], \tag{3.6}$$

for $\diamondsuit \in \{\mathsf{TV}, \mathsf{C}\}$, with the convention that the inner infimum and supremum are $+\infty$ and $-\infty$, respectively, when the robustness set is empty at θ. For each fixed θ, we refer to the optimization over P as the *inner problem* and to the optimization over θ as the *outer problem*. Section 4 shows how to solve the inner problems in (3.6) exactly despite their infinite-dimensional nature.

The sensitivity exercise we propose in this work reports $[\underline{\mathsf{b}}_{\diamondsuit}(\delta), \overline{\mathsf{b}}_{\diamondsuit}(\delta)]$ for $\delta \in [0,1]$. At $\delta = 0$ the

interval collapses to the baseline value of the estimand $\mathbb{E}_{P_*}[g(Z;\theta)]$.[11] As δ grows, the sensitivity bounds widen monotonically because the robustness sets are nested. The smallest δ at which the interval contains a given value, zero say, thus answers the question "how large does the share of the population for which the baseline assumption fails need to be to overturn the conclusion?" Section 5 formalizes this answer through summary measures.

We now map the three leading applications into this framework.

Example 2.1, continued. We can map the randomized experiment with attrition into the framework as follows. First, there is no structural parameter, so $\mathsf{d}_m = \mathsf{d}_\theta = 0$ and $\Theta = \{0\}$. The random vector of the framework is the treated potential outcome, that is $Z = Y(1)$ and $\mathcal{Z} = \mathcal{Y}$. The target estimand is $g(y;\theta) = y$. The baseline is the outcome distribution of the treated respondents, $P_* = P_{\mathsf{T}}$, which is identified by the data and must therefore be estimated, whereas the distribution being perturbed is the complier distribution P_{C}. In this notation, MCAR in the form (2.4) is exactly the identifying equality between the distribution of Z, here P_{C}, and the baseline P_{T}. The mixture decomposition (2.3) is the mixture constraint (3.3) with $\kappa = \pi$, the complier share in (2.2), with P playing the role of the complier distribution and R that of the always-observed distribution. The relevant robustness set is thus the total variation one in (3.4), and in fact $\mathcal{P}_{\mathsf{TV}}(\theta;\delta) = \mathcal{P}_{\mathsf{RCT}}(\delta)$ for every $\delta \in [0,1]$. In line with the interpretation given above, δ is the share of the treated respondents for whom MCAR may fail.[12] With these choices, the bounds in (3.6) are bounds on the complier mean,

$$\underline{\mathsf{b}}_{\mathsf{TV}}(\delta) = \inf_{P \in \mathcal{P}_{\mathsf{RCT}}(\delta)} \mathbb{E}_P[Y] \qquad \text{and} \qquad \overline{\mathsf{b}}_{\mathsf{TV}}(\delta) = \sup_{P \in \mathcal{P}_{\mathsf{RCT}}(\delta)} \mathbb{E}_P[Y].$$

The complier mean is the only quantity in (2.5) that the data do not identify, and $\tau(\cdot)$ is affine and decreasing in it, so the sensitivity bounds of Example 2.1 are

$$\underline{\tau}(\delta) = \frac{\mu_{\mathsf{T}} - \pi\,\overline{\mathsf{b}}_{\mathsf{TV}}(\delta)}{1-\pi} - \mu^{\mathsf{AO}}(0) \qquad \text{and} \qquad \overline{\tau}(\delta) = \frac{\mu_{\mathsf{T}} - \pi\,\underline{\mathsf{b}}_{\mathsf{TV}}(\delta)}{1-\pi} - \mu^{\mathsf{AO}}(0),$$

and both reduce to the MCAR estimand τ_{MCAR} at $\delta = 0$. Two features of this application are worth anticipating. First, the mixing weight π is estimated alongside the baseline. Second, when outcomes are continuously distributed, $\underline{\tau}(\delta)$ and $\overline{\tau}(\delta)$ admit a closed form, given in Theorem SA-34 of the supplementary materials, and $[\underline{\tau}(1), \overline{\tau}(1)]$ coincides with the sharp nonparametric bounds for τ_0 of Proposition 1a in Lee (2009). When pre-treatment covariates are added to tighten the bounds, there is no closed form for $\delta \in (0,1)$, whereas at $\delta = 1$ the sharp nonparametric bounds of Proposition 1b in Lee (2009) and Lemma 3 in Semenova (2025) are recovered (see Section SA6.1.1 of the supplementary materials). ♦

Example 2.2, continued. In this case, the random vector Z is the vector of preference shocks, that is $Z = U$ and so $\mathcal{Z} = \mathbb{R}^{J+1}$. The structural parameter collects the systematic components of

[11] This statement presumes that there exists a $\theta_* \in \Theta$ for which the baseline satisfies the moment conditions in (3.2), so that at $\delta = 0$ the baseline is the only distribution in the robustness set at the only feasible parameter value, a common assumption in the majority of the studies involving (generalized) method of moments estimators (see, e.g., Section 2.2.3 in Newey and McFadden, 1994).

[12] Equivalently, δ can be interpreted as the share of compliers or always-observed units for whom MCAR may fail.

match utility, $\theta = \vartheta$ and $\Theta \subseteq \mathbb{R}^{2J}$, the estimand is the marital college premium, $g = f$, and the moment function is $m = s$, so that $\mathsf{d}_m = 4J+2$ and $\vec{\rho}_0 = (\varrho, 0_{2(J+1)})$. The first $2J$ components of $\vec{\rho}_0$ are the observed marriage probabilities, which are estimated from the data, whereas the remaining $2(J+1)$ components are the normalizations. The baseline is the Gumbel specification of the shocks, $P_* = \widetilde{P}$, which is known rather than estimated. Unlike Example 2.1, the baseline is not itself a mixture of two groups, so the relevant robustness set is the contamination one: (2.7) is exactly (3.5) with the choices above, that is, $\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta) = \mathcal{P}_{\mathsf{C}}(\vartheta;\delta)$ for every $\vartheta \in \Theta$ and $\delta \in [0,1]$. Its two groups are the women whose shocks are Gumbel, a share $1-\delta$ of the population, and the women whose shocks are drawn from an arbitrary distribution R, a share δ, so that δ is the share of women for whom the Gumbel assumption may fail. Finally, the sensitivity bounds are $\underline{\mathsf{b}}_{\mathsf{C}}(\delta) = \underline{\mathsf{k}}(\delta)$ and $\overline{\mathsf{b}}_{\mathsf{C}}(\delta) = \overline{\mathsf{k}}(\delta)$ for every $\delta \in [0,1]$. In contrast with Example 2.1, these bounds admit no closed form, and the dual problems of Section 4 are the route through which we compute them in Section 6.2. ♦

Example 2.3, continued. Contrary to Examples 2.1 and 2.2, the target of this sensitivity exercise is the ratio $\widehat{\beta}(P)$ of two expectations, which is not of the form (3.1). Its numerator and denominator, however, are both affine in the candidate assignment process P, and this is what we leverage to cast the problem in the desired form. Indeed, for $b \in \mathbb{R}$ we have $\widehat{\beta}(P) = b$ if and only if $G_y(P) - b\,G_x(P) = 0$, and

$$G_y(P) - b\,G_x(P) = g_b(\check{v}) - \mathbb{E}_P[g_b(v)], \qquad g_b(\cdot) := g_y(\cdot) - b\,g_x(\cdot),$$

in view of (2.10). In words, the recentered estimator attains the value b under the candidate assignment process P exactly when $\mathbb{E}_P[g_b(v)]$ equals the realized value $g_b(\check{v})$. Once b is fixed, g_b is a known function on $\mathcal{V}$, and the exercise fits the framework upon setting $Z = v$, $\mathcal{Z} = \mathcal{V}$, $g(v;\theta) = g_b(v)$, $\Theta = \{0\}$, $\mathsf{d}_m = \mathsf{d}_\theta = 0$, and $P_* = \widetilde{P}$, the assignment process of the exogenous shocks postulated by the econometrician. With these choices, the robustness set (2.11) is the contamination set (3.5), that is $\mathcal{P}_{\mathsf{FI}}(\delta) = \mathcal{P}_{\mathsf{C}}(\theta;\delta)$ for every $\delta \in [0,1]$, where we use that g_b is bounded, so that every $P \in \Delta(\mathcal{V})$ belongs to $\mathcal{P}(\mathcal{Z})$. The interpretation of δ is that the exogenous shocks are drawn a fraction $(1-\delta)$ of the time from the postulated assignment process $\widetilde{P}$ and a fraction δ of the time from an arbitrary assignment process, so that δ is the share of the population for which the assumption that the shocks are assigned by $\widetilde{P}$ may fail. Write $\underline{\mathsf{b}}_{\mathsf{C}}(\delta; g_b)$ and $\overline{\mathsf{b}}_{\mathsf{C}}(\delta; g_b)$ for the bounds in (3.6) obtained with $g = g_b$, making the dependence on b explicit. Because $\mathcal{P}_{\mathsf{FI}}(\delta)$ is convex and $P \mapsto \mathbb{E}_P[g_b(v)]$ is linear, the values of $\mathbb{E}_P[g_b(v)]$ generated by the robustness set form an interval whose endpoints are these two bounds. It follows that every value b taken by $\widehat{\beta}(P)$ over $\mathcal{P}_{\mathsf{FI}}(\delta)$ belongs to the set

$$\mathsf{B}(\delta) := \Big\{ b \in \mathbb{R} : \ \underline{\mathsf{b}}_{\mathsf{C}}(\delta; g_b) \le g_b(\check{v}) \le \overline{\mathsf{b}}_{\mathsf{C}}(\delta; g_b) \Big\}.$$

The sensitivity bounds of Example 2.3 are therefore the endpoints of $\mathsf{B}(\delta)$,

$$\underline{\beta}(\delta) = \inf \mathsf{B}(\delta) \qquad \text{and} \qquad \overline{\beta}(\delta) = \sup \mathsf{B}(\delta).$$

Computing them thus amounts to solving the two inner problems in (3.6) for each b. ♦

4 Implementation

The inner problems in (3.6) optimize over an infinite-dimensional space of distributions and, outside special cases such as the RCT with attrition of Example 2.1, admit no closed form. In this section we show that they can nonetheless be solved exactly through a convex program in $\mathsf{d}_m + 2$ variables which, solved over a grid of values for δ, traces the thick orange lines of Figure 1. Section 4.1 derives the finite-dimensional dual of the inner problem for each robustness set of Section 3 and states conditions under which its value coincides with that of the primal in (3.6). Sections 4.2–4.3 then turn to finite samples: the former introduces plug-in estimators of the bounds, and the latter derives their limit distribution and proposes a resampling procedure to form valid confidence intervals.

4.1 Duality

Fix $\theta \in \Theta$ and $\Diamond \in \{\mathsf{TV}, \mathsf{C}\}$, and write

$$\overline{V}^{\Diamond}_{\mathsf{P}}(\theta;\delta) := \sup_{P \in \mathcal{P}_{\Diamond}(\theta;\delta)} \mathbb{E}_P[g(Z;\theta)] \tag{4.1}$$

for the value of the inner problem in the upper bound of (3.6). The lower bound is recovered by applying every result of this section to $-g$ in place of g, so we focus on the upper bound throughout.

The estimand and the moment restrictions are linear in P, and both robustness sets are convex, because the mixture representations in (3.4) and (3.5) are preserved under convex combinations and the total variation distance is jointly convex in its two arguments. The inner problem is therefore a convex program over probability distributions, and we construct its dual with elementary tools from convex analysis, following the logic of the duality results for distributionally robust optimization in Kuhn, Shafiee and Wiesemann (2025).[13] The appeal of the dual is that it looks at the problem from the point of view of its constraints rather than that of the candidate distribution, and thereby reduces its dimensionality from that of the space of distributions to $\mathsf{d}_m + 2$.

For $\theta \in \Theta$ and $\delta \in (0,1]$, the optimization problems dual to (4.1) with robustness sets (3.4) and (3.5) are respectively

$$\begin{aligned}\overline{V}^{\mathsf{TV}}_{\mathsf{D}}(\theta;\delta) := \inf_{\substack{(\zeta,\eta,\lambda)\in \\ \mathbb{R}\times\mathbb{R}_+\times\mathbb{R}^{\mathsf{d}_m}}} \Big\{ & \zeta + \eta(1-\kappa)\delta + \lambda^\top \vec{\rho}_0 \\ & + \mathbb{E}_{P_*}\big[\mathsf{r}_{\mathsf{TV}}\big(g(Z;\theta) - \lambda^\top m(Z;\theta) - \zeta,\ \eta\big)\big]\Big\}\end{aligned} \tag{4.2}$$

and

$$\begin{aligned}\overline{V}^{\mathsf{C}}_{\mathsf{D}}(\theta;\delta) := \inf_{\substack{(\zeta,\eta,\lambda)\in \\ \mathbb{R}\times\mathbb{R}_+\times\mathbb{R}^{\mathsf{d}_m}}} & \Big\{\zeta + \eta\delta + \lambda^\top \vec{\rho}_0 + \mathbb{E}_{P_*}\big[\mathsf{r}_{\mathsf{C}}\big(g(Z;\theta) - \lambda^\top m(Z;\theta) - \zeta,\ \eta\big)\big]\Big\} \\ \text{s.to} \quad & \zeta + \tfrac{\eta}{2} \ \geq\ \sup_{z\in\mathcal{Z}} \big\{g(z;\theta) - \lambda^\top m(z;\theta)\big\},\end{aligned} \tag{4.3}$$

[13] For further details, we refer the interested reader to Section SA2 of the supplementary materials. For a textbook treatment of the relevant convex analysis tools, see Rockafellar (1970, Sections 6, 12 and 31) and Rockafellar (1974).

where, for $x \in \mathbb{R}$ and $\eta \geq 0$,

$$\mathsf{r}_{\mathsf{TV}}(x,\eta) := \max\Big\{-\frac{\eta}{2},\ x,\ \frac{x}{\kappa} - \frac{(1-\kappa)\,\eta}{2\kappa}\Big\},$$
$$\mathsf{r}_{\mathsf{C}}(x,\eta) := \begin{cases} \max\Big\{(1-\delta)\,x - \frac{\delta\,\eta}{2},\ x\Big\}, & x \leq \eta/2, \\ +\infty, & x > \eta/2. \end{cases} \tag{4.4}$$

Some comments on the two duals are in order. First, both programs have $\mathsf{d}_m + 2$ variables, one per restriction of the primal problem other than the mixture restriction: $\zeta \in \mathbb{R}$ is the multiplier associated with the constraint that P is a probability distribution; $\eta \in \mathbb{R}_+$ to the budget constraint, which limits the share of the population whose distribution may depart from the baseline; $\lambda \in \mathbb{R}^{\mathsf{d}_m}$ to the moment equalities. Second, the three expressions inside the maximum defining r_{TV} are the values of the associated Lagrangian when a unit of the baseline population is removed from the group described by P, kept at its baseline weight, or assigned the largest weight compatible with the mixture restriction; the two expressions in r_{C} are its values when the unit is scaled down to the share $1-\delta$ or kept at its baseline weight. Third, (4.3) carries one additional constraint for each point of $\mathcal{Z}$, because candidate distributions in $\mathcal{P}_{\mathsf{C}}(\theta;\delta)$ may place mass outside the support of P_*, and the constraints make such relocations unprofitable. No such constraint appears in (4.2), because every $P \in \mathcal{P}_{\mathsf{TV}}(\theta;\delta)$ is absolutely continuous with respect to P_*, the mixture restriction forcing $\kappa P \leq P_*$. Fourth, r_{C} embeds an implicit constraint of its own. Since $\mathsf{r}_{\mathsf{C}}(x,\eta) = +\infty$ whenever $x > \eta/2$, the objective of (4.3) is finite only if $\zeta + \eta/2 \geq g(z;\theta) - \lambda^\top m(z;\theta)$ for P_*-almost every $z \in \mathcal{Z}$. The pointwise constraint in (4.3) imposes the same inequality at every $z \in \mathcal{Z}$ and therefore implies the implicit one. Proposition 1 below gives a formal condition under which the converse also holds, so that the pointwise constraint is redundant and can be deleted.

The next theorem shows that solving (4.2) and (4.3) yields the same value as (4.1). Toward this goal, define the sets of attainable moments

$$\Gamma_{\mathsf{TV}}(\theta;\delta) := \big\{\mathbb{E}_P[m(Z;\theta)] :\ P \in \mathcal{P}(\mathcal{Z}),\ \mathsf{TV}(P \,\|\, R) \leq \delta,\ P_* = \kappa P + (1-\kappa)R \text{ for some } R \in \mathcal{P}(\mathcal{Z})\big\},$$
$$\Gamma_{\mathsf{C}}(\theta;\delta) := \big\{\mathbb{E}_P[m(Z;\theta)] :\ P \in \mathcal{P}(\mathcal{Z}),\ P = (1-\delta)P_* + \delta R \text{ for some } R \in \mathcal{P}(\mathcal{Z})\big\},$$

which collect the moment vectors generated by the distributions satisfying every restriction defining the corresponding robustness set other than the moment equalities in (3.2).

Theorem 1 (Strong duality)**.** *Let $\theta \in \Theta$ and $\delta \in (0,1]$.*

(i) If $\vec{\rho}_0 \in \mathrm{ri}(\Gamma_{\mathsf{TV}}(\theta;\delta))$, then $\overline{V}_{\mathsf{P}}^{\mathsf{TV}}(\theta;\delta) = \overline{V}_{\mathsf{D}}^{\mathsf{TV}}(\theta;\delta)$.

(ii) If $\sup_{z\in\mathcal{Z}} |g(z;\theta)| < \infty$ and $\vec{\rho}_0 \in \mathrm{ri}(\Gamma_{\mathsf{C}}(\theta;\delta))$, then $\overline{V}_{\mathsf{P}}^{\mathsf{C}}(\theta;\delta) = \overline{V}_{\mathsf{D}}^{\mathsf{C}}(\theta;\delta)$.

The condition $\vec{\rho}_0 \in \mathrm{ri}(\Gamma_{\Diamond}(\theta;\delta))$ is a Slater condition. In words, it asks that the moment restrictions remain attainable, within the same budget δ, after a small perturbation of the target $\vec{\rho}_0$ in every direction along which the attainable moments vary. The relative interior formulation accommodates models whose moment components satisfy exact affine relations. When no moment restrictions are

present, the condition holds under the conventions of Section 3. The boundedness of g in part (ii) is the price of letting the candidate distributions place mass everywhere in $\mathcal{Z}$, even where P_* does not.

The pointwise constraint, potentially an infinite collection of inequalities, is what makes (4.3) more burdensome to compute than (4.2), and we now give a condition under which it can be dropped without affecting the value of the program. For $\theta \in \Theta$ and $\delta \in (0,1]$ define

$$\overline{V}_{\mathsf{P}}^{\mathsf{C},\ll}(\theta;\delta) := \sup_{P \in \mathcal{P}_{\mathsf{C}}^{\ll}(\theta;\delta)} \mathbb{E}_P[g(Z;\theta)], \qquad \mathcal{P}_{\mathsf{C}}^{\ll}(\theta;\delta) := \mathcal{P}_{\mathsf{C}}(\theta;\delta) \cap \{P : P \ll P_*\},$$

for the value of the inner problem over the contamination robustness set restricted to distributions dominated by the baseline, write $\overline{V}_{\mathsf{D}}^{\mathsf{C},\ll}(\theta;\delta)$ for the infimum of the objective of (4.3) over all of $\mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}^{\mathsf{d}_m}$, that is, for the value of (4.3) with the pointwise constraint deleted, and let

$$\Gamma_{\mathsf{C}}^{\ll}(\theta;\delta) := \left\{ \mathbb{E}_P[m(Z;\theta)] : \ P \in \mathcal{P}(\mathcal{Z}),\ P \ll P_*,\ P = (1-\delta)P_* + \delta R,\ R \in \mathcal{P}(\mathcal{Z}) \right\}$$

be the attainable moment set of the restricted problem, a subset of $\Gamma_{\mathsf{C}}(\theta;\delta)$. The inequalities $\overline{V}_{\mathsf{P}}^{\mathsf{C},\ll}(\theta;\delta) \le \overline{V}_{\mathsf{P}}^{\mathsf{C}}(\theta;\delta)$ and $\overline{V}_{\mathsf{D}}^{\mathsf{C},\ll}(\theta;\delta) \le \overline{V}_{\mathsf{D}}^{\mathsf{C}}(\theta;\delta)$ hold in general, the first because $\mathcal{P}_{\mathsf{C}}^{\ll}(\theta;\delta) \subseteq \mathcal{P}_{\mathsf{C}}(\theta;\delta)$ and the second because deleting a constraint enlarges the feasible set.

Proposition 1 (Deleting the pointwise constraint)**.** *Let $\theta \in \Theta, \delta \in (0,1]$, $\vec{\rho}_0 \in \mathrm{ri}(\Gamma_{\mathsf{C}}^{\ll}(\theta;\delta))$, $\sup_{z\in\mathcal{Z}} |g(z;\theta)| < \infty$, and, for every $\lambda \in \mathbb{R}^{\mathsf{d}_m}$,*

$$\sup_{z\in\mathcal{Z}} \{g(z;\theta) - \lambda^\top m(z;\theta)\} = P_*\text{-}\operatorname*{ess\,sup}_{z\in\mathcal{Z}} \{g(z;\theta) - \lambda^\top m(z;\theta)\}, \tag{4.5}$$

then $\overline{V}_{\mathsf{P}}^{\mathsf{C}}(\theta;\delta) = \overline{V}_{\mathsf{P}}^{\mathsf{C},\ll}(\theta;\delta) = \overline{V}_{\mathsf{D}}^{\mathsf{C},\ll}(\theta;\delta)$.

Proposition 1 contains two results. The first equality states that, under its regularity conditions, removing the distributions $P \not\ll P_*$ from $\mathcal{P}_{\mathsf{C}}(\theta;\delta)$ does not change the value of (4.1). The second states that this value is delivered by the dual (4.3) without the pointwise constraint. Condition (4.5) holds, for example, whenever $\mathrm{supp}(P_*) = \mathcal{Z}$ and $z \mapsto g(z;\theta) - \lambda^\top m(z;\theta)$ is lower semicontinuous for every λ, as in the formula instrument setting of Example 2.3, and also for the matching indicators of Example 2.2, which are not continuous.

4.2 Estimation

Having recast the inner problems as tractable convex programs, we now turn to estimating the bounds, that is, the thick orange lines in Figure 1. Toward this goal, we introduce some notation. Partition the moment function as $m = (m_{\mathsf{e}}^\top, m_{\mathsf{k}}^\top)^\top$, where the target of the moment function $m_{\mathsf{e}} : \mathcal{Z} \times \Theta \to \mathbb{R}^{\mathsf{d}_{\mathsf{e}}}, \mathsf{d}_{\mathsf{e}} \in \mathbb{N}_0$ is estimated and that of $m_{\mathsf{k}} : \mathcal{Z} \times \Theta \to \mathbb{R}^{\mathsf{d}_{\mathsf{k}}}, \mathsf{d}_{\mathsf{k}} \in \mathbb{N}_0$ is known and normalized to zero. Throughout this section, $\vec{\rho}_0 := (\rho_0^\top, 0_{\mathsf{d}_{\mathsf{k}}}^\top)^\top$ where $\rho_0 \in \mathbb{R}^{\mathsf{d}_{\mathsf{e}}}$ denotes the moment target to be estimated. Accordingly, partition the multipliers as $\lambda = (\lambda_{\mathsf{e}}^\top, \lambda_{\mathsf{k}}^\top)^\top$ with $\lambda_{\mathsf{e}} \in \mathbb{R}^{\mathsf{d}_{\mathsf{e}}}$ and $\lambda_{\mathsf{k}} \in \mathbb{R}^{\mathsf{d}_{\mathsf{k}}}$. Throughout, we let $\rho \in \mathbb{R}^{\mathsf{d}_{\mathsf{e}}}$, $P \in \Delta(\mathcal{Z})$, and $\xi := (\zeta, \eta, \lambda)$ collect the dual

variables. To treat the two robustness sets at once, let $c_{\mathsf{TV}} := 1-\kappa$ and $c_{\mathsf{C}} := 1$ be the factors that multiply the term $\eta\delta$ in (4.2) and (4.3), respectively. For $\Diamond \in \{\mathsf{TV}, \mathsf{C}\}$, we write the objective of the corresponding dual problem as

$$\overline{\mathsf{J}}^{\Diamond}(\xi, \theta; \rho, P) := \zeta + c_{\Diamond}\,\eta\delta + \lambda_{\mathsf{e}}^{\top}\rho + P\overline{\ell}^{\Diamond}_{\xi,\theta}, \quad \overline{\ell}^{\Diamond}_{\xi,\theta}(z) := \mathsf{r}_{\Diamond}\big(g(z;\theta) - \lambda^{\top} m(z;\theta) - \zeta,\ \eta\big),$$

with $\mathsf{r}_{\Diamond}$ as in (4.4), and its feasible set as $\overline{\mathcal{F}}^{\mathsf{TV}}(\theta) := \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}^{\mathsf{d}_m}$ and

$$\overline{\mathcal{F}}^{\mathsf{C}}(\theta) := \Big\{\xi \in \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}^{\mathsf{d}_m} :\ \zeta + \tfrac{\eta}{2} \geq \sup_{z \in \mathcal{Z}} \big\{g(z;\theta) - \lambda^{\top} m(z;\theta)\big\}\Big\},$$

so that $\overline{\mathcal{F}}^{\mathsf{C}}(\theta)$ is the set cut out by the pointwise constraint of (4.3), on which the integrand r_{C} is finite, whereas we recall that (4.2) has no such constraint. Symmetrically, let $\underline{\mathsf{J}}^{\Diamond}$, $\underline{\ell}^{\Diamond}_{\xi,\theta}$, and $\underline{\mathcal{F}}^{\Diamond}(\theta)$ be defined as $\overline{\mathsf{J}}^{\Diamond}$, $\overline{\ell}^{\Diamond}_{\xi,\theta}$, and $\overline{\mathcal{F}}^{\Diamond}(\theta)$ with $-g$ in place of g, so that, in particular, $\underline{\mathcal{F}}^{\mathsf{TV}}(\theta) = \overline{\mathcal{F}}^{\mathsf{TV}}(\theta)$. Further, let $\mathcal{P}_{\Diamond}(\theta; \delta, \rho, P)$ denote the robustness set (3.4) or (3.5) with $(\vec{\rho}_0, P_*)$ replaced by $((\rho^{\top}, 0_{\mathsf{d}_k}^{\top})^{\top}, P)$, so that $\mathcal{P}_{\Diamond}(\theta; \delta)$ equals $\mathcal{P}_{\Diamond}(\theta; \delta, \rho_0, P_*)$. In line with the convention following (3.6), we define the criterion functions

$$\overline{K}^{\Diamond}_{\delta}(\theta; \rho, P) := \begin{cases} \inf\limits_{\xi \in \overline{\mathcal{F}}^{\Diamond}(\theta)} \overline{\mathsf{J}}^{\Diamond}(\xi, \theta; \rho, P) & \text{if } \mathcal{P}_{\Diamond}(\theta; \delta, \rho, P) \neq \emptyset, \\ -\infty & \text{otherwise,} \end{cases} \tag{4.6}$$

and

$$\underline{K}^{\Diamond}_{\delta}(\theta; \rho, P) := \begin{cases} -\inf\limits_{\xi \in \underline{\mathcal{F}}^{\Diamond}(\theta)} \underline{\mathsf{J}}^{\Diamond}(\xi, \theta; \rho, P) & \text{if } \mathcal{P}_{\Diamond}(\theta; \delta, \rho, P) \neq \emptyset, \\ +\infty & \text{otherwise.} \end{cases} \tag{4.7}$$

Finally, let

$$\mathsf{D}_{\Diamond}(\theta; \rho, P) := \inf_{Q \in \mathcal{P}_{\Diamond}(\theta; \delta, \rho, P)} \mathsf{TV}(Q \,\|\, P) \text{ and } \Theta^{\Diamond}_{\delta}(\rho, P) := \big\{\theta \in \Theta :\ \mathsf{D}_{\Diamond}(\theta; \rho, P) < c_{\Diamond}\,\delta\big\}$$

be the smallest distance from P to the robustness set and the set of strictly feasible parameter values. The sensitivity bounds can thus be written as

$$\underline{\mathsf{b}}_{\Diamond}(\delta, \rho, P) := \inf_{\theta \in \Theta^{\Diamond}_{\delta}(\rho, P)} \underline{K}^{\Diamond}_{\delta}(\theta; \rho, P) \quad \text{and} \quad \overline{\mathsf{b}}_{\Diamond}(\delta, \rho, P) := \sup_{\theta \in \Theta^{\Diamond}_{\delta}(\rho, P)} \overline{K}^{\Diamond}_{\delta}(\theta; \rho, P),$$

for $\Diamond \in \{\mathsf{TV}, \mathsf{C}\}$. At the true inputs (ρ_0, P_*), these coincide with the sensitivity bounds $\underline{\mathsf{b}}_{\Diamond}(\delta)$ and $\overline{\mathsf{b}}_{\Diamond}(\delta)$ in (3.6). The plug-in estimators replace the two inputs by $\widehat{\rho}_n$, an estimator of ρ_0, and by the empirical measure $\widehat{P}_n$ obtained from an i.i.d. sample from P_*, which we recall is an identified feature of P_0, the true distribution of Z. Formally, the plug-in estimators are $\widehat{\underline{\mathsf{b}}}_{\Diamond,n}(\delta) := \underline{\mathsf{b}}_{\Diamond}(\delta, \widehat{\rho}_n, \widehat{P}_n)$ and $\widehat{\overline{\mathsf{b}}}_{\Diamond,n}(\delta) := \overline{\mathsf{b}}_{\Diamond}(\delta, \widehat{\rho}_n, \widehat{P}_n)$. When the baseline is instead known but the moment target is estimated, as in Example 2.2, the estimators are simply $\underline{\mathsf{b}}_{\Diamond}(\delta, \widehat{\rho}_n, P_*)$ and $\overline{\mathsf{b}}_{\Diamond}(\delta, \widehat{\rho}_n, P_*)$.

Remark 1 (Over-identified models)**.** When the model is over-identified, i.e., when there are more moments (d_m) than parameters (d_θ), there might not exist a $\theta \in \Theta$ for which $\mathbb{E}_{P_*}[m(Z;\theta)]$ equals the estimated target $(\widehat{\rho}_n^{\top}, 0_{\mathsf{d}_k}^{\top})^{\top}$. We refer to Section 3.3 in Christensen and Connault (2023) for two alternative solutions to handle this case. ♣

4.3 Large Sample Properties

Our large sample theory rests on two key properties of (4.6), which hold for both robustness sets: (i) the objective $\overline{\mathsf{J}}^{\Diamond}$ is affine in (ρ, P); (ii) the feasible set $\overline{\mathcal{F}}^{\Diamond}(\theta)$ does not depend on (ρ, P). As a consequence, the map $(\rho, P) \mapsto \overline{K}^{\Diamond}_{\delta}(\theta; \rho, P)$ is concave with supergradient $(\lambda_{\mathsf{e}}, \overline{\ell}^{\Diamond}_{\xi,\theta})$ at any dual minimizer, and derivatives of the bounds can be read off the dual solutions by an envelope argument. The same two properties hold for (4.7), so that $(\rho, P) \mapsto \underline{K}^{\Diamond}_{\delta}(\theta; \rho, P)$ is convex with subgradient $-(\lambda_{\mathsf{e}}, \underline{\ell}^{\Diamond}_{\xi,\theta})$.

The theory of this section provides large sample guarantees when both the moment target ρ_0 and the baseline P_* are estimated from the data. Remark 2 below explains how it simplifies when P_* is known, as in Example 2.2. In terms of Figure 1, we now turn to the dashed orange lines denoting the confidence intervals of the bounds.

Before stating the conditions needed by the large sample theory, we introduce some useful quantities. For $\Diamond \in \{\mathsf{TV}, \mathsf{C}\}$, let

$$\overline{\Theta}^{\Diamond}_{\delta}(\rho, P) := \operatorname*{arg\,sup}_{\theta \in \Theta} \overline{K}^{\Diamond}_{\delta}(\theta\,; \rho, P) \quad \text{and} \quad \overline{\Xi}^{\Diamond}_{\delta}(\theta; \rho, P) := \operatorname*{arg\,inf}_{\xi \in \overline{\mathcal{F}}^{\Diamond}(\theta)} \overline{\mathsf{J}}^{\Diamond}(\xi, \theta; \rho, P)$$

be the set of outer optimizers of the dual problem and the set of optimal dual multipliers at (θ, ρ, P), respectively, and

$$\overline{\Lambda}^{\Diamond}_{\delta}(\theta; \rho, P) := \big\{\lambda_{\mathsf{e}} \in \mathbb{R}^{\mathsf{d}_{\mathsf{e}}} : (\zeta, \eta, \lambda) \in \overline{\Xi}^{\Diamond}_{\delta}(\theta; \rho, P) \text{ with } \lambda = (\lambda_{\mathsf{e}}, \lambda_{\mathsf{k}}) \text{ for some } (\zeta, \eta, \lambda_{\mathsf{k}})\big\}$$

be the projection of $\overline{\Xi}^{\Diamond}_{\delta}(\theta; \rho, P)$ onto the space of multipliers λ_{e}. The quantities $\underline{\Theta}^{\Diamond}_{\delta}$, $\underline{\Xi}^{\Diamond}_{\delta}$, and $\underline{\Lambda}^{\Diamond}_{\delta}$ are defined in the same way from the program (4.7). The following assumption lists all the regularity conditions needed for the large sample theory.

Assumption LS. *We observe an i.i.d. sample $\{Z_i\}_{i=1}^n$ from P_*. Furthermore, the following conditions hold:*

(i) (Compactness.) $\mathcal{Z}$ and Θ are nonempty compact sets.

(ii) (Continuity.) The functions $g(z;\theta)$ and $m(z;\theta)$ are jointly continuous on $\mathcal{Z} \times \Theta$ and there exists a constant $L < \infty$ such that for all $\theta, \theta' \in \Theta$

$$\sup_{z \in \mathcal{Z}} |g(z;\theta) - g(z;\theta')| \leq L\|\theta - \theta'\| \quad \text{and} \quad \sup_{z \in \mathcal{Z}} \|m(z;\theta) - m(z;\theta')\| \leq L\|\theta - \theta'\|.$$

(iii) (Feasibility.) $\Theta^{\Diamond}_{\delta}(\rho_0, P_*) \neq \emptyset$.

(iv) (Slater condition.) For each $\theta \in \underline{\Theta}^{\Diamond}_{\delta}(\rho_0, P_*) \cup \overline{\Theta}^{\Diamond}_{\delta}(\rho_0, P_*)$, $\vec{\rho}_0 \in \operatorname{int}\big(\Gamma_{\Diamond}(\theta; \delta)\big)$.

(v) (Asymptotic linearity.) We have $\widehat{\rho}_n = \rho_0 + \widehat{P}_n\psi + o_{\mathbb{P}}(n^{-1/2})$, with influence function $\psi : \mathcal{Z} \to \mathbb{R}^{\mathsf{d}_{\mathsf{e}}}$ satisfying $P_*\psi = 0$ and $P_*\|\psi\|^2 < \infty$.

(vi) (Lower hemicontinuity.) The correspondences $\theta \mapsto \underline{\Xi}^{\Diamond}_{\delta}(\theta; \rho_0, P_*)$ and $\theta \mapsto \overline{\Xi}^{\Diamond}_{\delta}(\theta; \rho_0, P_*)$ are lower hemicontinuous at each $\theta \in \underline{\Theta}^{\Diamond}_{\delta}(\rho_0, P_*)$ and each $\theta \in \overline{\Theta}^{\Diamond}_{\delta}(\rho_0, P_*)$, respectively.

Assumption LS(i) greatly simplifies the integrability conditions and, together with Assumptions LS(ii) and LS(iii), guarantees the existence of solutions to the outer problem.[14] In Assumption LS(ii), joint continuity is used, together with compactness, to control the supremum in the pointwise constraint of $\overline{\mathcal{F}}^{\mathsf{C}}(\theta)$ and for the existence of solutions to the outer problem, whereas Lipschitz continuity is needed to invoke an appropriate functional central limit theorem and is vacuous when there is no structural parameter, as in Examples 2.1 and 2.3. Assumption LS(iii) ensures that the criterion functions are not identically $\pm\infty$. Assumption LS(iv) is the Slater-type condition discussed after Theorem 1, stated with the interior rather than the relative interior, which rules out exact affine relations among the moment components. It also guarantees that the dual minimizer sets $\underline{\Xi}_\delta^\Diamond(\theta;\rho_0,P_*)$ and $\overline{\Xi}_\delta^\Diamond(\theta;\rho_0,P_*)$ are nonempty, compact, and convex (Lemma SA-24 of the supplementary materials), so that the multiplier sets $\underline{\Lambda}_\delta^\Diamond, \overline{\Lambda}_\delta^\Diamond$ are well defined. Assumption LS(v) requires the estimator of ρ_0 to be asymptotically linear, so that the joint law of $\sqrt{n}(\widehat{\rho}_n-\rho_0)$ and of the empirical process $\mathbb{G}_n := \sqrt{n}(\widehat{P}_n - P_*)$ is amenable to the delta method used below. Finally, Assumption LS(vi) is used to establish the directional differentiability of the value function. Under Assumption LS the plug-in estimators for the bounds are consistent.

Theorem 2 (Consistency)**.** *Let Assumption LS hold for $\Diamond \in \{\mathsf{TV},\mathsf{C}\}$. Then*

$$\underline{\mathsf{b}}_\Diamond(\delta,\widehat{\rho}_n,\widehat{P}_n) \xrightarrow{\mathbb{P}} \underline{\mathsf{b}}_\Diamond(\delta,\rho_0,P_*) \qquad \textit{and} \qquad \overline{\mathsf{b}}_\Diamond(\delta,\widehat{\rho}_n,\widehat{P}_n) \xrightarrow{\mathbb{P}} \overline{\mathsf{b}}_\Diamond(\delta,\rho_0,P_*).$$

To establish a distributional approximation for the bounds, we rely on a delta method for maps that are only Hadamard directionally differentiable, that is, whose derivatives are continuous but not necessarily linear.[15] The reason is that the bounds are values of optimization problems. The maps $(\rho,P)\mapsto(\underline{\mathsf{b}},\overline{\mathsf{b}})(\delta,\rho,P)$ are Hadamard differentiable when these programs have a single optimizer, but generally only directionally so when they have several, as Christensen and Connault (2023, Section 6.1) already noted for the case of known P_*. We therefore use the delta method for Hadamard directionally differentiable functionals (see, e.g., Shapiro, 1991; Dümbgen, 1993). The next result first derives the directional derivative of the bounds at (ρ_0,P_*) and then establishes their distributional approximation.

Theorem 3 (Directional differentiability and distributional approximation)**.** *Let Assumption LS hold for $\Diamond \in \{\mathsf{TV},\mathsf{C}\}$. Then the maps $(\rho,P)\mapsto(\underline{\mathsf{b}}_\Diamond,\overline{\mathsf{b}}_\Diamond)(\delta,\rho,P)$ are Hadamard directionally differ-*

[14] We stress that some of the motivating examples do not satisfy compactness of $\mathcal{Z}$ in Assumption LS(i). In Example 2.1 the outcome space $\mathcal{Y}$ is a closed, possibly unbounded, subset of $\mathbb{R}$, whereas in Example 2.2, $\mathcal{Z}=\mathbb{R}^{J+1}$. For Example 2.1, Proposition SA-35 of the supplementary materials exploits the closed form of the bounds to replace compactness by a finite second moment of the outcome and a density positive at the trimming quantiles, and it also accounts for the estimation of the complier share. For Example 2.2, relaxing compactness of $\mathcal{Z}$ would require assuming a coercivity condition on the moment function, which we leave for future work. In the same example, the matching indicators in s_1 also violate the continuity requirement of Assumption LS(ii), but only on the set of ties, which is null under the Gumbel baseline, so that this requirement can be relaxed as well.

[15] For a formal definition of Hadamard differentiability and Hadamard directional differentiability, see Section 2.2 in Fang and Santos (2019).

entiable at (ρ_0, P_*)*, with*

$$d\underline{\mathsf{b}}_\diamond(\delta,\rho_0,P_*)[h,\gamma] = \min_{\theta^*\in\underline{\Theta}^\diamond_\delta(\rho_0,P_*)} \max_{\xi^*\in\underline{\Xi}^\diamond_\delta(\theta^*;\rho_0,P_*)} \big\{ -(\lambda^*_{\mathsf{e}})^\top h - \gamma\, \underline{\ell}^\diamond_{\xi^*,\theta^*}\big\},$$

$$d\overline{\mathsf{b}}_\diamond(\delta,\rho_0,P_*)[h,\gamma] = \max_{\theta^*\in\overline{\Theta}^\diamond_\delta(\rho_0,P_*)} \min_{\xi^*\in\overline{\Xi}^\diamond_\delta(\theta^*;\rho_0,P_*)} \big\{(\lambda^*_{\mathsf{e}})^\top h + \gamma\, \overline{\ell}^\diamond_{\xi^*,\theta^*}\big\},$$

and so

$$\sqrt{n}\begin{pmatrix} \underline{\mathsf{b}}_\diamond(\delta,\widehat{\rho}_n,\widehat{P}_n) - \underline{\mathsf{b}}_\diamond(\delta,\rho_0,P_*) \\ \overline{\mathsf{b}}_\diamond(\delta,\widehat{\rho}_n,\widehat{P}_n) - \overline{\mathsf{b}}_\diamond(\delta,\rho_0,P_*)\end{pmatrix} \rightsquigarrow \begin{pmatrix} d\underline{\mathsf{b}}_\diamond(\delta,\rho_0,P_*)[Z_\psi,\mathbb{G}] \\ d\overline{\mathsf{b}}_\diamond(\delta,\rho_0,P_*)[Z_\psi,\mathbb{G}]\end{pmatrix},$$

where $\mathbb{G}$ *is a* P_**-Brownian bridge and* $Z_\psi = \mathbb{G}\psi$*, the pair* $(Z_\psi,\mathbb{G})$ *being jointly Gaussian with* $\mathbb{E}[\mathbb{G}f\, Z_\psi] = P_*(f\psi)$*.*

The novelty of Theorem 3 lies in the fact that the limiting distribution carries the term $\mathbb{G}\overline{\ell}^\diamond_{\xi^*,\theta^*}$, which is also correlated with Z_ψ as both are functionals of the same sample. When the sets of outer maximizers and dual minimizers are singletons, the directional derivatives become linear and the limit is Gaussian with variance

$$\sigma^2_* = \mathbb{V}_{P_*}\big((\lambda^*_{\mathsf{e}})^\top\psi + \overline{\ell}^\diamond_{\xi^*,\theta^*}(Z)\big),$$

where the two terms in the variance capture the uncertainty due to the estimation of the moment target ρ_0 and of the baseline P_*, respectively, and their covariance reflects that both are estimated from the same sample.

Outside this singleton case the derivatives in Theorem 3 are, in general, nonlinear, and the standard bootstrap is invalid (Fang and Santos, 2019, Theorem 3.1). Indeed, the standard bootstrap may be interpreted as relying on a (numerical) derivative that consistently estimates the directional derivative only when the latter is linear (see Remark 3.5 in Fang and Santos, 2019). Accordingly, in the same spirit as Christensen and Connault (2023), we specialize the resampling procedure proposed in Fang and Santos (2019) to our case. Concretely, we estimate the directional derivative as

$$\widehat{d\underline{\mathsf{b}}}_{\diamond,n}(\delta)[h,\gamma] := \sup_{\theta\in\widehat{\underline{\Theta}}^\diamond_{\delta,n}} \inf_{\xi\in\underline{\Xi}^\diamond_\delta(\theta;\widehat{\rho}_n,\widehat{P}_n)} \big\{\lambda_{\mathsf{e}}^\top h + \gamma\, \overline{\ell}^\diamond_{\xi,\theta}\big\},$$

$$\widehat{\underline{\Theta}}^\diamond_{\delta,n} := \Big\{\theta\in\Theta^\diamond_\delta(\widehat{\rho}_n,\widehat{P}_n) : \overline{K}^\diamond_\delta(\theta;\widehat{\rho}_n,\widehat{P}_n) \geq \widehat{\underline{\mathsf{b}}}_{\diamond,n}(\delta) - \widehat{\nu}_n\sqrt{\tfrac{\log n}{n}}\Big\}$$

where $\widehat{\nu}_n \geq 0$ is a tuning sequence that is bounded and bounded away from zero in probability. The lower-bound counterparts $\widehat{\underline{\Theta}}^\diamond_{\delta,n}$ and $\widehat{d\underline{\mathsf{b}}}_{\diamond,n}(\delta)$ are defined analogously, with $-g$ in place of g. Following Christensen and Connault (2023, Section 6.2), a convenient data-driven choice of $\widehat{\nu}_n$ is the smallest diagonal entry of the covariance matrix of $m(Z;\widehat{\theta}_n)$ under $\widehat{P}_n$, with $\widehat{\theta}_n$ an approximate maximizer of the estimated outer problem, which requires no simulation. Algorithm 1 details all the steps of the resampling procedure, whose validity is proven in the supplementary materials under a strengthening of Assumption LS. In particular, we show that the bootstrap statistics $\widehat{T}^{*,b,\text{low}}_n$ and $\widehat{T}^{*,b,\text{up}}_n$ of Algorithm 1 are consistent for their population counterparts in Theorem 3, and the confidence sets they deliver have asymptotic coverage at least $1-\alpha$ whenever the distribution functions of those limits are continuous and strictly increasing at the quantiles used.

Algorithm 1 Confidence sets for the sensitivity bounds

1: Compute $\widehat{\rho}_n$, $\widehat{P}_n$, the bounds $\underline{\widehat{\mathsf{b}}}_{\Diamond,n}(\delta), \overline{\widehat{\mathsf{b}}}_{\Diamond,n}(\delta)$, the near-argmax sets $\underline{\widehat{\Theta}}^{\Diamond}_{\delta,n}, \overline{\widehat{\Theta}}^{\Diamond}_{\delta,n}$, and the dual minimizer sets $\overline{\Xi}^{\Diamond}_{\delta}(\theta; \widehat{\rho}_n, \widehat{P}_n)$ and $\underline{\Xi}^{\Diamond}_{\delta}(\theta; \widehat{\rho}_n, \widehat{P}_n)$.
2: **for** $b = 1, \ldots, B$ **do**
3: Draw an i.i.d. sample $\{Z_i^{*,b}\}_{i=1}^n$ from $\widehat{P}_n$ and form $\widehat{P}_n^{*,b}$ and $\widehat{\rho}_n^{*,b}$.
4: Set $\widehat{T}_n^{*,b,\mathrm{up}} := \widehat{d\overline{\mathsf{b}}}_{\Diamond,n}(\delta)\big[\sqrt{n}(\widehat{\rho}_n^{*,b} - \widehat{\rho}_n),\ \sqrt{n}(\widehat{P}_n^{*,b} - \widehat{P}_n)\big]$ and the analog $\widehat{T}_n^{*,b,\mathrm{low}}$.
5: **end for**
6: For $u \in (0,1)$, let $\widehat{\mathfrak{c}}_u^{\mathrm{low}}$ and $\widehat{\mathfrak{c}}_u^{\mathrm{up}}$ be the empirical u-quantiles of $\{\widehat{T}_n^{*,b,\mathrm{low}}\}_{b=1}^B$ and $\{\widehat{T}_n^{*,b,\mathrm{up}}\}_{b=1}^B$, respectively. Valid lower and upper $100(1-\alpha)\%$ confidence sets are

$$[\underline{\widehat{\mathsf{b}}}_{\Diamond,n}(\delta) - \widehat{\mathfrak{c}}_{1-\alpha}^{\mathrm{low}}/\sqrt{n}, +\infty) \quad \text{and} \quad (-\infty, \overline{\widehat{\mathsf{b}}}_{\Diamond,n}(\delta) - \widehat{\mathfrak{c}}_{\alpha}^{\mathrm{up}}/\sqrt{n}].$$

Remark 2 (Known baseline)**.** When the baseline is known, the same argument with P held fixed at P_* yields consistency, while the directional derivative becomes

$$d\overline{\mathsf{b}}_{\Diamond}(\delta, \rho_0, P_*)[h] = \max_{\theta^* \in \overline{\Theta}^{\Diamond}_{\delta}(\rho_0, P_*)} \ \min_{\lambda^* \in \overline{\Lambda}^{\Diamond}_{\delta}(\theta^*; \rho_0, P_*)} (\lambda^*)^\top h,$$

which is identical to that of Christensen and Connault (2023, Theorem 6.2) even though total variation sits outside the class of divergences considered there. In this case, the Lipschitz requirement of Assumption LS(ii) is not needed, and the lower hemicontinuity of Assumption LS(vi) may be imposed on the multiplier projections $\underline{\Lambda}^{\Diamond}_{\delta}, \overline{\Lambda}^{\Diamond}_{\delta}$ alone, since the derivative then sees the dual optimizer only through them. Moreover, since expectations against P_* are computed exactly, the data enter only through $\widehat{\rho}_n$. Accordingly, Assumption LS(v) may be weakened to the requirement that $\sqrt{n}(\widehat{\rho}_n - \rho_0) \rightsquigarrow \mathsf{N}(0, \Sigma)$ for a finite $\mathsf{d_e} \times \mathsf{d_e}$ matrix Σ, with $\widehat{\rho}_n$ computed from whatever sample identifies ρ_0, which need not be a sample of Z. In Example 2.2, Z collects latent preference shocks that are never observed while ρ_0 is estimated from data on realized matches. Then, Step 3 of Algorithm 1 draws only a bootstrap replicate of $\widehat{\rho}_n$ and the resulting procedure coincides with the algorithm of Christensen and Connault (2023, Section 6.2) (see Remark SA-11 of the supplementary materials for formal statements). ♣

5 How to Use the Sensitivity Bounds?

This section explains how the output of our sensitivity framework is meant to be used. In terms of Figure 1, Section 5.1 studies the right endpoint of the horizontal axis, where at $\delta = 1$ the bounds coincide with the exact nonparametric sharp bounds, and Section 5.2 introduces some reportable summary measures.

5.1 Exact Nonparametric Sharp Bounds

Besides admitting the population share interpretation, measuring the relaxation of the distributional assumption through shares has an important byproduct: the sensitivity parameter never

needs to exceed one. At $\delta = 1$, the total variation constraint in (3.4) holds for every pair of groups, because the total variation distance never exceeds one, and the mixture representation in (3.5) reads $P = R$, so that it places no restriction on P. What remains binding at $\delta = 1$ are the restrictions that the model itself imposes, namely that Z takes values in $\mathcal{Z}$, that the moment restrictions (3.2) hold at θ, and, for the total variation robustness set, the mixture constraint (3.3). Therefore, the sensitivity bounds at $\delta = 1$ coincide with the nonparametric sharp bounds on the estimand (3.1). The next theorem formalizes this intuition. Toward this goal, we define

$$\mathcal{E}(\theta) := \big\{P \in \mathcal{P}(\mathcal{Z}) : \ \mathbb{E}_P[m(Z;\theta)] = \vec{\rho}_0\big\},$$

the set of distributions compatible with the moment restrictions at θ,

$$\mathcal{D}_{\mathsf{TV}}(P_*) := \big\{P \in \mathcal{P}(\mathcal{Z}) : \ \kappa P \leq P_*\big\} \qquad \text{and} \qquad \mathcal{D}_{\mathsf{C}}(P_*) := \mathcal{P}(\mathcal{Z}).$$

Theorem 4 (Nonparametric sharp bounds)**.** *Let* $\Diamond \in \{\mathsf{TV}, \mathsf{C}\}$*. Then, for* $\theta \in \Theta$*,*

$$\mathcal{P}_\Diamond(\theta; 1) = \mathcal{D}_\Diamond(P_*) \cap \mathcal{E}(\theta).$$

Consequently, the sensitivity bounds (3.6) *at* $\delta = 1$ *satisfy*

$$\underline{\mathsf{b}}_\Diamond(1) = \inf_{\theta\in\Theta} \inf_{P\in\mathcal{D}_\Diamond(P_*)\cap\mathcal{E}(\theta)} \mathbb{E}_P[g(Z;\theta)], \quad \overline{\mathsf{b}}_\Diamond(1) = \sup_{\theta\in\Theta} \sup_{P\in\mathcal{D}_\Diamond(P_*)\cap\mathcal{E}(\theta)} \mathbb{E}_P[g(Z;\theta)].$$

Theorem 4 follows directly from the definitions of the two robustness sets. In words, $\underline{\mathsf{b}}_{\mathsf{C}}(1)$ and $\overline{\mathsf{b}}_{\mathsf{C}}(1)$ are the infimum and the supremum of $\mathbb{E}_P[g(Z;\theta)]$ over all $\theta \in \Theta$ and all $P \in \mathcal{E}(\theta)$, that is, over every distribution on $\mathcal{Z}$ that satisfies the moment restrictions at some parameter value, whereas $\underline{\mathsf{b}}_{\mathsf{TV}}(1)$ and $\overline{\mathsf{b}}_{\mathsf{TV}}(1)$ further require the baseline to admit the mixture representation (3.3). In both cases the bounds are sharp, meaning that no distribution compatible with the maintained restrictions generates a value of the estimand outside $[\underline{\mathsf{b}}_\Diamond(1), \overline{\mathsf{b}}_\Diamond(1)]$, so that neither bound can be tightened without imposing restrictions beyond those maintained.

Theorem 4 goes beyond sensitivity analysis as it makes the framework a device for computing nonparametric sharp bounds. In some problems those bounds have known closed-form expressions. For instance, in Example 2.1 the interval $[\underline{\mathsf{b}}_{\mathsf{TV}}(1), \overline{\mathsf{b}}_{\mathsf{TV}}(1)]$ delivers the trimming bounds of Lee (2009) when outcomes are continuously distributed (see Theorem SA-34 of the supplementary materials). Similarly, in the missing data problem of Section SA6.4.3 of the supplementary materials, where a share p of the population is observed and its outcome law is the baseline, the contamination bounds $[\underline{\mathsf{b}}_{\mathsf{C}}(1-p), \overline{\mathsf{b}}_{\mathsf{C}}(1-p)]$ at a budget equal to the missing share coincide with the sharp nonparametric bounds of Manski (1994, 2003). In other problems, such as the marital college premium of Example 2.2, no closed form is available, and the dual problems of Section 4 return the sharp nonparametric bounds numerically.

5.2 Summary Measures

Throughout this subsection we let $\Diamond \in \{\mathsf{TV}, \mathsf{C}\}$ and suppress it from the notation, so that $\underline{\mathsf{b}}(\delta)$ and $\overline{\mathsf{b}}(\delta)$ denote the bounds in (3.6). We measure the robustness of a conclusion against a *reference*

value that the analyst wants to rule out, such as zero for the sign of the estimand or a policy relevant threshold. For simplicity, we set it to zero and take the effect to be positive at the baseline, $\underline{\mathsf{b}}(0) > 0$, the opposite case being symmetric. The estimated counterparts we report in Section 6 are plug-in estimators of the four summary measures defined below. Section SA5 of the supplementary materials studies the four measures and establishes the large sample properties of the estimated certified breakdown budget. Its formal results are stated for the total variation robustness set without structural parameters, which is the case of the attrition application of Section 6.1, but the properties below hold for the contamination robustness set by the same arguments.

Breakdown and certified breakdown. The *breakdown budget* is the smallest budget at which the sensitivity bounds include the reference value, that is,

$$\delta_{\mathsf{b}} \equiv \delta_{\mathsf{b}}(0) := \inf\big\{\delta \in [0,1] : \ \underline{\mathsf{b}}(\delta) \leq 0\big\}, \tag{5.1}$$

with the convention that the infimum over an empty set equals 1. The breakdown point goes back to the robust statistics literature (Hampel, 1968; Donoho and Huber, 1983) and has been adopted in econometrics to quantify how much a maintained assumption can be relaxed before a conclusion is overturned (see, among many, Horowitz and Manski, 1995; Kline and Santos, 2013; Masten and Poirier, 2020; Ober-Reynolds, 2026). By the population share interpretation of Section 3, no departure from the identifying assumption involving less than a δ_{b} share of the population can overturn the conclusion.

The breakdown budget captures solely misspecification uncertainty. To account for sampling uncertainty as well, for $\alpha \in (0,1)$ and a sample size n held fixed, we define the *certified breakdown budget*

$$\delta_{\mathsf{cb}}(\alpha; n) := \inf\Big\{\delta \in [0,1] : \ \underline{\mathsf{b}}(\delta) - \frac{\mathfrak{c}_{1-\alpha/2}\,\sigma(\delta)}{\sqrt{n}} \leq 0\Big\}, \tag{5.2}$$

where $\sigma(\delta)$ is the asymptotic standard deviation of the plug-in estimator $\widehat{\underline{\mathsf{b}}}_n(\delta)$ and $\mathfrak{c}_{1-\alpha/2}$ is the $(1-\alpha/2)$-th quantile of the standard normal distribution.[16] The critical value is two-sided because the exercise does not commit in advance to the sign of $\underline{\mathsf{b}}(\delta)$. We say that a study *certifies* robustness at a budget δ and level α when its conclusion survives both misspecification of size δ and sampling uncertainty at level $1-\alpha$. Similar measures appear in Cinelli and Hazlett (2020) for unobserved confounding and in Rambachan and Roth (2023) for violations of parallel trends.

By construction, $\delta_{\mathsf{cb}}(\alpha; n) \leq \delta_{\mathsf{b}}$. Moreover, $\delta_{\mathsf{cb}}(\alpha; n)$ is nondecreasing in n and converges to δ_{b} as $n \to \infty$, and, under the conditions of Proposition SA-32 of the supplementary materials, its sample counterpart is consistent for δ_{b}. Sampling uncertainty thus vanishes but robustness does not, because δ_{b} is a feature of the design alone. Reporting the pair $(\widehat{\delta}_{\mathsf{b}}, \widehat{\delta}_{\mathsf{cb}})$ therefore separates fragility by design (a small breakdown budget) from fragility due to imprecision (a large gap between the two). The two budgets can also rank studies differently: a decision maker weighing whether to fund a second round of data collection should prefer the study with the larger breakdown budget,

[16] This definition presumes that the problems defining $\underline{\mathsf{b}}(\delta)$ have a unique optimizer, so that the limit of $\sqrt{n}(\widehat{\underline{\mathsf{b}}}_n(\delta) - \underline{\mathsf{b}}(\delta))$ is Gaussian by Theorem 3; see Remark SA-13 of the supplementary materials for the attrition application and the general case.

whereas one who has to act on the evidence in hand should prefer the study with the larger certified budget.

The shadow price of robustness. We hold the structural parameter fixed and suppress it from the notation. Under appropriate regularity conditions, the bound paths are finite, monotone, and concave ($\overline{\mathsf{b}}$) and convex ($\underline{\mathsf{b}}$), respectively, on the range of budgets at which the robustness set is nonempty, so that each path is differentiable at all but at most countably many budgets. At every such budget we define the *shadow price of robustness* as

$$\overline{\eta}(\delta) := \overline{\mathsf{b}}'(\delta) \qquad \text{and} \qquad \underline{\eta}(\delta) := -\underline{\mathsf{b}}'(\delta).$$

Both shadow prices are nonnegative and nonincreasing in the budget, so that the first unit of misspecification is the most damaging. For the total variation robustness set, the budget enters (4.2) only through the additive term $\eta(1-\kappa)\delta$, so that η prices the total variation constraint as a Lagrange multiplier prices a resource constraint, and $\overline{\eta}(\delta) = (1-\kappa)\,\eta^*$ wherever the optimal multiplier η^* is unique (Proposition SA-31 of the supplementary materials).

Unlike the breakdown budget, which concerns the sign of an effect, the shadow price also speaks to its magnitude. Suppose that a decision maker is unwilling to commit to a single budget and instead holds beliefs μ on $[0,1]$, with distribution function F_μ. Then, provided that $\underline{\mathsf{b}}$ is finite and continuous on $[0,1]$, she could evaluate the design by integrating the worst case sensitivity bound $\underline{\mathsf{b}}(\delta)$ over her beliefs, that is

$$\mathbb{E}_\mu\left[\underline{\mathsf{b}}(\delta)\right] = \underline{\mathsf{b}}(0) - \int_0^1 \underline{\eta}(s)\left(1 - F_\mu(s)\right) \mathrm{d}s,$$

so that the price of the s-th unit of misspecification is weighted by the probability she assigns to a violation at least that large.

The certification frontier. For a budget $\delta < \delta_{\mathsf{b}}$, the *certification frontier*

$$n^\star(\delta;\alpha) := \frac{\mathfrak{c}^2_{1-\alpha/2}\,\sigma^2(\delta)}{\underline{\mathsf{b}}(\delta)^2} \tag{5.3}$$

is the sample size a study would need in order to certify robustness at budget δ and level α. Under the conditions of Proposition SA-33 of the supplementary materials, its semi-elasticity with respect to δ behaves as $2/(\delta_{\mathsf{b}}-\delta)$ as $\delta \to \delta_{\mathsf{b}}$, so that each additional percentage point of certified robustness becomes increasingly expensive in observations. More data, however, cannot make a study more robust: as $n \to \infty$ the certified budget converges to δ_{b} and no further, because misspecification is of fixed size while sampling uncertainty shrinks at rate $n^{-1/2}$. Put differently, the frontier prices certification, not robustness itself.

6 Empirical Applications

We present three empirical applications. The first revisits the online tutoring program of Carlana and La Ferrara (2025), a randomized experiment in which attrition is moderate but strongly differential across arms. The second revisits the marital college premium application of Christensen and

Connault (2023) and focuses on interpretation and computational cost. The third turns to formula instruments: it assesses the sensitivity of the estimated elasticity of Borusyak and Hull (2023) to the postulated assignment process of the shocks. The supplementary materials contain two further exercises on attrition. Section SA7.1 revisits the transport subsidy experiment of Christensen, Nino and Osman (2026), in which attrition is severe but nearly balanced across arms, and Section SA7.2 runs the same sensitivity exercise on every eligible trial in RCT-Bench, the public collection of randomized clinical trials assembled by Shao, Lyu, Yu and Wang (2026), which currently comprises 125 trials, asking how often attrition threatens the conclusions of medical experiments in practice.

6.1 Attrition in Randomized Experiments

We revisit the Tutoring Online Program experiment of Carlana and La Ferrara (2025), maintaining the setup of Example 2.1 throughout.

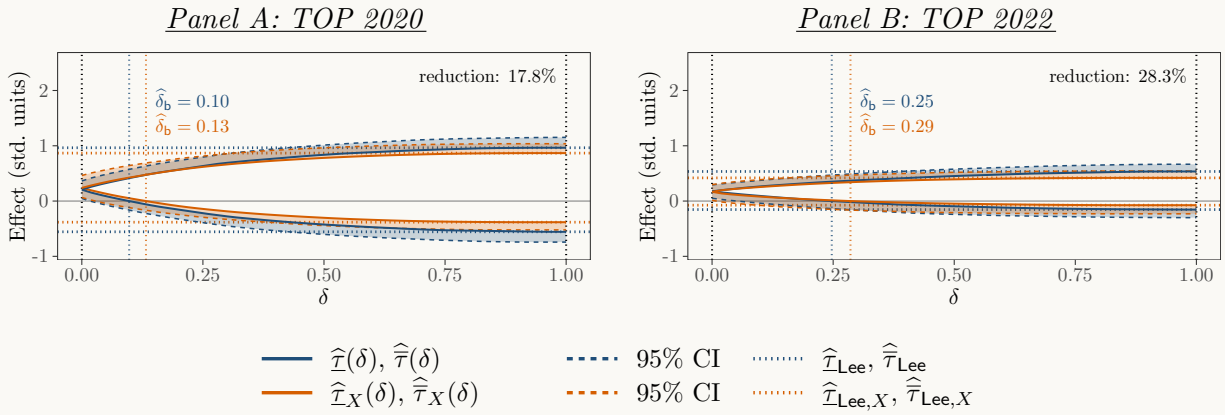
Setup. Carlana and La Ferrara (2025) evaluate the Tutoring Online Program (TOP), which offered free individual online tutoring by university student volunteers to disadvantaged Italian middle schoolers. The experiment ran in two waves, with individual-level randomization in each: TOP 2020, during the COVID-19 school closures (530 treated, 529 control), and TOP 2022, during regular in-person schooling (607 treated, 530 control). The outcome is a standardized mathematics performance index that is observed only for test takers (i.e., respondents). Table 1 shows that attrition varies across waves and arms: the response rates of the treated and control arms are 87.9% and 46.3% in 2020 and 92.3% and 72.3% in 2022, implying estimated complier shares $\widehat{\pi}$ of 0.47 and 0.22 in the 2020 and 2022 samples, respectively. Our sensitivity exercise asks how much of the experimental conclusion survives departures from ignorable attrition. In addition, our framework allows us to compare and rank the two waves in terms of how robust their conclusions are.

Table 1: *Attrition and sample size by wave in the Tutoring Online Program.*

	Treatment		*Control*		
Wave	n	Attrition	n	Attrition	$\widehat{\pi}$
TOP 2020	530	12.1%	529	53.7%	0.47
TOP 2022	607	7.7%	530	27.7%	0.22

Results. Figure 2 reports the bounds for TOP 2020 (Panel A) and TOP 2022 (Panel B). In TOP 2020, the bounds start at $\delta = 0$ from the difference in means among respondents, which corresponds to an estimated 0.207 standard deviations, and at $\delta = 1$ reach the Lee bounds, $[-0.556, 0.964]$. The sign of the effect survives at most 10% of the compliers departing from the MCAR assumption. Once sampling uncertainty is taken into account, the sign is certified at the 5% level only up to $\widehat{\delta}_{\mathsf{cb}} = 0.017$. We do not report the certified breakdown budgets in Figure 2 to avoid cluttering the panels. Conditioning on a handful of pre-treatment covariates tightens the bounds at every budget, shrinking their width at $\delta = 1$ by 18%, and increases the breakdown budget to 0.13 and

Figure 2: TV *sensitivity bounds in the Tutoring Online Program.*



Notes: Computed from the replication data of Carlana and La Ferrara (2025). Each panel plots the plug-in estimators of $\underline{\tau}(\delta)$ and $\overline{\tau}(\delta)$ described in Section 4.2 over $\delta \in \{0, 0.02, \ldots, 1\}$, without and with covariates, together with the bounds of Lee (2009) and the outer 95% confidence bounds from 1,000 iterations of the bootstrap of Algorithm 1. The vertical dotted lines mark the estimated breakdown budgets $\hat{\delta}_{\mathsf{b}}$ without and with covariates, and the annotation in each panel reports the reduction in the width of the bounds at $\delta = 1$ achieved by the covariates. The covariate cells are the full interaction of indicators for grades six and seven, a median split of the baseline INVALSI math score, mother's high school completion, mother's white collar occupation, and single parent household.

the certified breakdown budget to 0.028.[17] In TOP 2022, the bounds start from a difference in means of 0.176 standard deviations and reach the Lee bounds, $[-0.154, 0.535]$. The sign of the effect survives at most 25% of the compliers departing from the MCAR assumption, and it is certified at the 5% level up to $\hat{\delta}_{\mathsf{cb}} = 0.051$. Covariates tighten the bounds at every budget above $\delta = 0.1$, shrinking their width at $\delta = 1$ by 28%, and increase the breakdown budget to 0.29. The sensitivity exercise thus ranks the two waves: TOP 2020 is the more fragile one, even though its difference in means among respondents is larger. In hindsight, this is not surprising, as TOP 2020 ran during the school closures and is also the wave with the larger response differential across arms. Neither the point estimates nor the Lee bounds, which do not sign the effect in either wave, can make this distinction.

Table 2 reports, for each specification, some of the summary measures of Section 5.2: the breakdown budget $\hat{\delta}_{\mathsf{b}}$, the certified breakdown budget $\hat{\delta}_{\mathsf{cb}}$ at the 5% level, the shadow price of robustness at the breakdown budget $\hat{\underline{\eta}}(\hat{\delta}_{\mathsf{b}})$, and the certification frontier evaluated five percentage points above the certified budget, together with its implied cost (see the table notes). Three comments are in order. First, in both waves the certified budget is far below the breakdown budget, so that most of the apparent fragility is due to sampling uncertainty, not to misspecification uncertainty. Second, certifying five additional percentage points of robustness requires roughly 8,000 additional students (\$80k per point) in TOP 2020, but only about 1,200 (\$12k per point) in TOP 2022, whose smaller complier share makes the bounds widen more slowly. The frontier thus answers the second round budgeting question raised in Section 5.2 and tells a decision maker funding a hypothetical second round where an additional dollar buys the most certified robustness. Third, the shadow price $\hat{\underline{\eta}}(\hat{\delta}_{\mathsf{b}})$

[17] We explain in detail how to add covariates in Section SA6.1.1 of the supplementary materials.

is minus the slope of the estimated lower bound in Figure 2 at the budget where it crosses zero. At the breakdown budget, each additional percentage point of compliers departing from MCAR lowers the worst case by 0.017 standard deviations in TOP 2020, against 0.005 in TOP 2022. As shadow prices are nonincreasing in the budget, the lower bound of TOP 2020 falls at least this fast at every smaller budget, while that of TOP 2022 never falls faster than its initial shadow price of 1.00. This is why TOP 2020 breaks down first despite its larger difference in means.

Table 2: *Summary measures in the Tutoring Online Program.*

	$\widehat{\delta}_{\mathsf{b}}$	$\widehat{\delta}_{\mathsf{cb}}$	$\widehat{\underline{\eta}}(\widehat{\delta}_{\mathsf{b}})$	n	$n^\star$	Δn	\$/pp
TOP 2020	0.098	0.017	1.74	1,059	9,048	+7,989	80k
TOP 2022	0.247	0.051	0.49	1,137	2,326	+1,189	12k

Notes: $\widehat{\delta}_{\mathsf{b}}$ is the plug-in breakdown budget (5.1) and $\widehat{\delta}_{\mathsf{cb}}$ the certified breakdown budget (5.2) at $\alpha = 0.05$, both without covariates; $\widehat{\underline{\eta}}(\widehat{\delta}_{\mathsf{b}})$ is the estimated shadow price of robustness at $\widehat{\delta}_{\mathsf{b}}$; n is the realized sample size and $n^\star$ the certification frontier (5.3) evaluated at $\widehat{\delta}_{\mathsf{cb}} + 0.05$, rounded up, with $\Delta n = n^\star - n$; \$/pp is the implied cost per percentage point of certified robustness at a marginal cost of \$50 per participant enrolled and surveyed (Puri and Rathinam, 2019).

Finally, we use the shadow price of robustness to offer a different comparison of the two waves. Consider a decision maker who has no information on the extent of the departure from MCAR and holds uniform beliefs μ on $[0,1]$, so that $F_\mu(s) = s$ and the evaluation of Section 5.2 reads $\mathbb{E}_\mu[\underline{\tau}(\delta)] = \underline{\tau}(0) - \int_0^1 \underline{\eta}(s)(1-s)\,\mathrm{d}s$. We compute the shadow prices as the slopes of the plug-in lower bound, without covariates, between consecutive budgets of the grid of Figure 2. In TOP 2020 the shadow price starts at 2.74, so that each of the first two percentage points of compliers departing from MCAR lowers the worst case by about 0.027 standard deviations, and the weighted integral equals 0.556, which gives $\mathbb{E}_\mu[\widehat{\underline{\tau}}(\delta)] = 0.207 - 0.556 = -0.349$. In TOP 2022 the shadow price starts at 1.00 and the integral equals 0.237, which gives $\mathbb{E}_\mu[\widehat{\underline{\tau}}(\delta)] = 0.176 - 0.237 = -0.061$. The decision maker thus ranks the waves in the opposite order to their differences in means because the larger point estimate of TOP 2020 is more than offset by a shadow price that starts almost three times as large.

6.2 Counterfactuals in Structural Models

We revisit one of the leading applications of CC23: the sensitivity of the marital college premium in the matching model of Chiappori et al. (2017) to the Gumbel specification of the taste shocks.

Setup. For illustrative purposes, we use the data for a single cohort instead of the 28 separate cohorts of CC23, namely cohort 14, composed of white men born in 1953 and white wives one year younger. There are $J = 5$ education groups (high school dropouts, high school graduates, some college, college graduates, and postgraduates), and we measure the premium between college graduate (b) and some college (b') women. We set $\Theta = [-12, 12]^{2J}$ and plug in the estimated matching frequencies for ϱ. At the Gumbel baseline the premium is -0.221. We obtain confidence bounds from the version of Algorithm 1 where P_* is known. Expectations over P_* are approximated

by 5,000 scrambled Halton draws, while the outer optimization problem is solved with ten random restarts per bound and warm starts along the δ-grid. We compare two sensitivity exercises using the same data, moments, and simulation draws. The first reports the bounds of CC23, which let the candidate distributions range over the robustness set

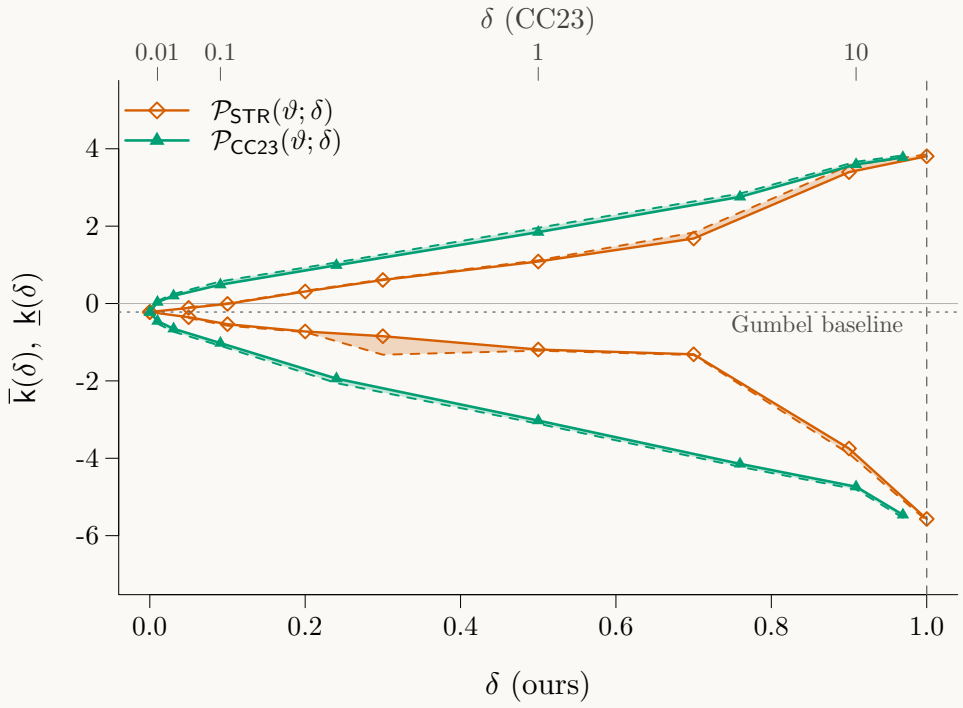
$$\mathcal{P}_{\mathsf{CC23}}(\vartheta;\delta) := \Big\{ P \in \mathcal{P}(\mathbb{R}^{J+1}) : P \ll P_*,\ \mathbb{E}_P[s(U;\vartheta)] = (\varrho, 0_{2(J+1)}), \\ \mathbb{E}_{P_*}\Big[\phi_{\mathsf{CC23}}\Big(\tfrac{\mathrm{d}P}{\mathrm{d}P_*}\Big)\Big] \le \delta \Big\}, \tag{6.1}$$

where

$$\phi_{\mathsf{CC23}}(x) := \begin{cases} x\log x - x + 1, & 0 \le x \le e, \\ \dfrac{1}{2e}(x-e)^2 + (x-e) + 1, & x > e. \end{cases}$$

In words, the divergence used to measure distances between a candidate P and the baseline P_* coincides with the Kullback–Leibler divergence for $x \le e$ and continues with a χ^2-type quadratic tail for $x > e$. CC23 adopt this hybrid because one of their regularity conditions fails for pure Kullback–Leibler at the Gumbel baseline. The second exercise delivers the bounds $\underline{\mathsf{k}}(\delta)$ and $\overline{\mathsf{k}}(\delta)$ of Example 2.2 over the robustness set $\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta)$ in (2.7), in which $P = (1-\delta)P_* + \delta R$, so that δ is the share of women drawing non-Gumbel shocks.

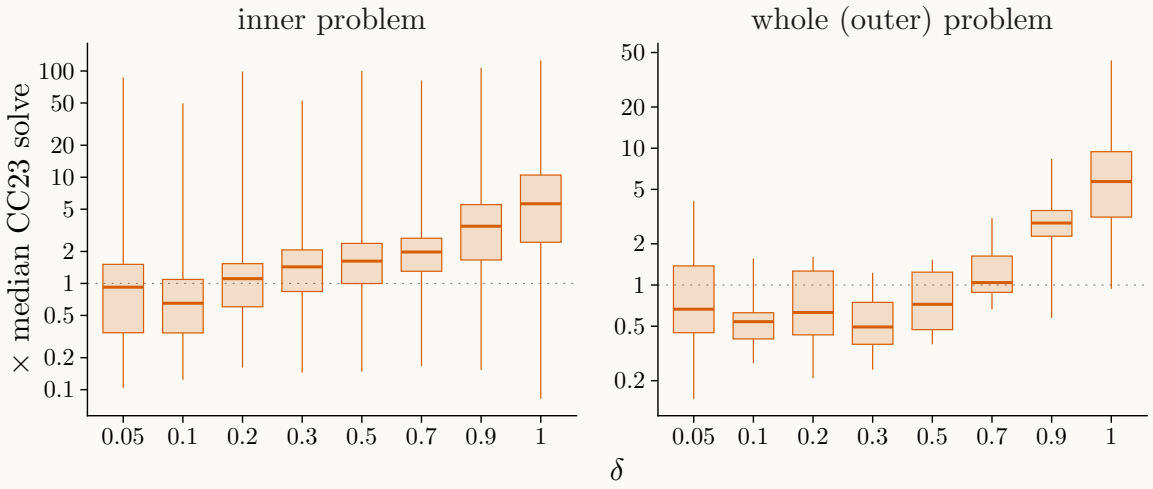
Figure 3: *Sensitivity of the marital college premium on a common axis.*



Notes: Computed from the cohort-14 marriage probabilities of white couples in the data of Chiappori et al. (2017), as prepared by the replication package of CC23. The figure plots the bounds $\underline{\mathsf{k}}(\delta)$ and $\overline{\mathsf{k}}(\delta)$ of Example 2.2 over the robustness set $\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta)$ of (2.7) (solid orange lines) and the bounds of CC23 over their robustness set $\mathcal{P}_{\mathsf{CC23}}(\vartheta;\delta)$ of (6.1) (solid green lines). Our bounds are plotted against the bottom x-axis, the budget δ of (2.7). The CC23 bounds are drawn at the horizontal position $\delta/(1+\delta)$, which maps their budget $\delta \in [0,\infty)$ onto $[0,1)$, and the top x-axis reports the original values of δ. Dashed lines are 95% confidence bounds from 1,000 draws of Algorithm 1.

Results. Figure 3 plots the two sets of bounds on a common axis. Three comments are in order. First, the sign of the premium is fragile. The upper bound $\overline{\mathsf{k}}(\delta)$ is still negative at $\delta = 0.1$, where it equals -0.010, but it is already positive at $\delta = 0.2$, where it equals 0.310. Hence, a share of women between a tenth and a fifth drawing non-Gumbel shocks suffices to overturn the sign of the baseline premium. Second, by Theorem 4, $\underline{\mathsf{k}}(1)$ and $\overline{\mathsf{k}}(1)$ are the sharp nonparametric bounds on the premium. As expected, the CC23 bounds approach the same envelope only in the large budget limit, without attaining it at any finite budget, coming very close at a budget of roughly 100, the largest value CC23 use in their application. Since our bounds attain the envelope, they can also be used as a diagnostic for how well a divergence-based approximation does at any given budget. Third, along the whole path sampling uncertainty is an order of magnitude smaller than misspecification uncertainty, suggesting that ambiguity, not noise, drives the width of the bounds. Figure 4 compares solve times. A CC23 inner solve takes on the order of a hundredth of a second at 5,000 draws, and the outer problem for one bound a few seconds (median between 0.4 and 5.3 seconds). Our programs solve in comparable time: up to $\delta = 0.7$, the median inner solve takes less than twice as long as that of CC23, while the median whole problem takes roughly the same time. Both ratios rise to between five and six at $\delta = 1$. Computing one bound over the eight-point budget grid takes about 3 minutes against 2.5 for CC23.

Figure 4: *Computational cost of the sensitivity bounds.*



Notes: Computed from the replication data of CC23, sharing the simulation draws of Figure 3. The panels report solve times for the bounds over the robustness set $\mathcal{P}_{\mathsf{STR}}(\vartheta;\delta)$ of (2.7), for the inner problem at a given ϑ (left panel) and for the whole problem (right panel), in multiples of the median solve time of the corresponding CC23 problem (dotted horizontal line) using $\mathcal{P}_{\mathsf{CC23}}(\vartheta;\delta)$ of (6.1), over the budget grid $\delta \in \{0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 0.9, 1\}$, displayed at equally spaced positions. All programs are solved with Knitro 15.1 in Julia 1.12 on an Apple M5 laptop.

6.3 Formula Instruments

We revisit one of the leading applications of BH23: the effect of market access growth on the growth of urban employment across Chinese prefectures between 2007 and 2016.

Setup. BH23 exploit the construction of the country's high-speed rail network. The market access of prefecture $i \in [n]$ in year $t \in \{2007, 2016\}$ is

$$\mathsf{MA}_{it} \equiv \mathsf{MA}_i(v_t) = \sum_{j \in [n]} \frac{\mathsf{Pop}_j}{\mathsf{Time}(v_t, \mathsf{Loc}_i, \mathsf{Loc}_j)},$$

where Pop_j is the population of prefecture j in the 2000 census and $\mathsf{Time}(\cdot)$ is the predicted travel time between the locations Loc_i and Loc_j given the status v_t of the railway lines in year t. The endogenous regressor is $x_i = \log \mathsf{MA}_{i,2016} - \log \mathsf{MA}_{i,2007}$, the outcome y_i is the growth of urban employment over the same period, and the sample consists of the $n = 275$ prefectures for which the employment outcome is recorded. Because market access grew most in the economic center of the country, a comparison with the periphery may be confounded by unobserved determinants of employment growth.

BH23 therefore use the timing of construction, rather than the location of the network, as the source of exogenous variation. The shocks $v \in \{0,1\}^K$ are the 2016 completion indicators of the $K = 149$ lines that were in service by 2016 (83) or completed or under construction by 2019 (66), and $\mathcal{V} = \{v^{(1)}, \ldots, v^{(S)}\}$ collects the $S = 1{,}999$ counterfactual networks obtained by permuting completion status within groups of lines that connect the same number of prefectures. The formula of Example 2.3 is $f_i(v'; w) = \log \mathsf{MA}_i(v') - \log \mathsf{MA}_{i,2007}$, where $w = \big((\mathsf{Loc}_j, \mathsf{Pop}_j)_{j \in [n]}, v_{2007}\big)$ is predetermined, so that $z_i = f_i(\check{v}; w) = x_i$ at the realized shocks $\check{v}$. Following BH23, $\widetilde{P}$ is the uniform distribution on $\mathcal{V}$, so that $\mu_i(\widetilde{P}) = S^{-1} \sum_{s=1}^{S} f_i(v^{(s)}; w)$. We consider both specifications of BH23, the first controlling for a constant only and the second adding distance to Beijing, latitude, and longitude. Recentering lowers the estimated elasticity from 0.232 to $\widehat{\beta}_* = 0.084$ without controls and from 0.133 to $\widehat{\beta}_* = 0.056$ with them. Condition (2.12) holds in both specifications. The recentered first stage at the postulated assignment process is $G_x(\widetilde{P}) = 9.92$ without controls and 9.24 with them, whereas $\sup_{v' \in \mathcal{V}} g_x(v') - \mathbb{E}_{\widetilde{P}}[g_x(v)]$ equals 8.34 and 5.66, respectively.

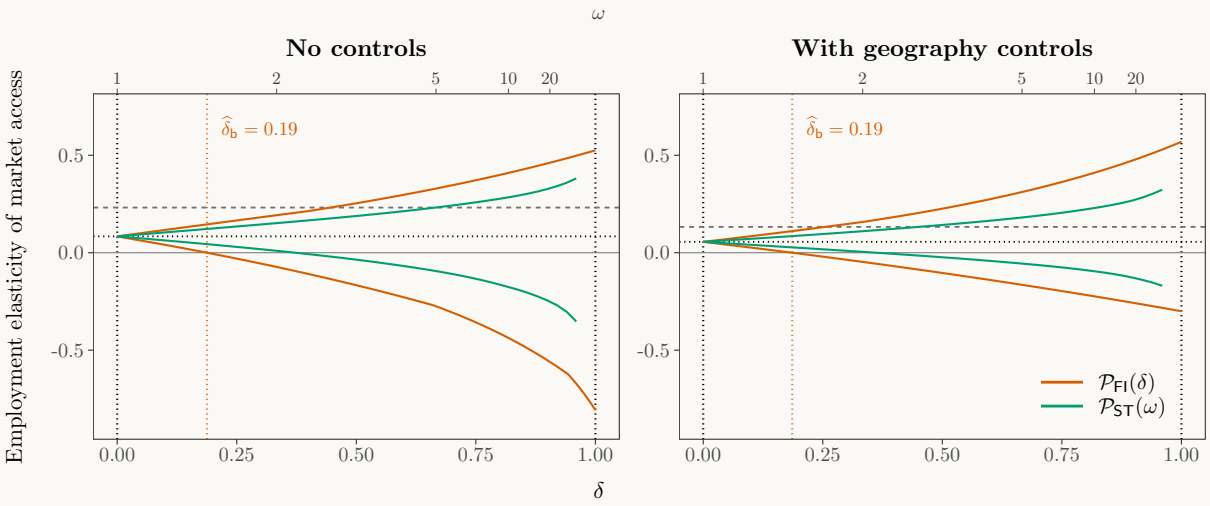
Figure 5 reports two types of bounds. The first are our sensitivity bounds $\underline{\beta}(\delta)$ and $\overline{\beta}(\delta)$ over the robustness set $\mathcal{P}_{\mathsf{FI}}(\delta)$ of (2.11) (Proposition SA-37 in the supplementary materials characterizes them and their computation). The second are the bounds of Sheng and Torgovitsky (2026) (ST26, henceforth), who ask the same question for this very application and vary the assignment process over what they call "the joint sensitivity set"

$$\mathcal{P}_{\mathsf{ST}}(\omega) := \Big\{ P \in \Delta(\mathcal{V}) : \; P \ll \widetilde{P}, \; \omega^{-1} \leq \mathrm{d}P / \mathrm{d}\widetilde{P} \leq \omega, \; \widetilde{P}\text{-a.s.} \Big\}, \tag{6.2}$$

for $\omega \geq 1$, which rescales the postulated probability of each configuration by a factor of at most ω in either direction. Like ours, and unlike their marginal sensitivity set, this set constrains the law of the configurations.

Results. Three features of this exercise are worth commenting on. First, the sign of the estimate is fragile: the sensitivity bounds contain zero from a budget of $\widehat{\delta}_{\mathsf{b}} = 0.19$ onwards, both without and with controls. In other words, a mechanism other than the postulated one governing roughly a fifth of the shock draws suffices for the estimated employment effect to lose its sign. Second, the sensitivity bounds include the unadjusted least squares estimate at $\delta = 0.45$ without controls, and

Figure 5: *Sensitivity bounds for the employment effect of market access growth.*



Notes: Computed from the replication data of BH23. Each panel plots the bounds $\underline{\beta}(\delta)$ and $\overline{\beta}(\delta)$ of Example 2.3 over the robustness set $\mathcal{P}_{\mathsf{FI}}(\delta)$ of (2.11) (solid orange lines), and the ST26 bounds over the set (6.2) (solid green lines), the latter drawn against the rescaling $(\omega - 1)/(\omega + 1)$ of their budget, whose original values are reported on the top horizontal axis. The dotted horizontal line is the recentered estimate $\widehat{\beta}_*$ of (2.9) and the dashed horizontal line the unadjusted OLS estimate. The orange vertical dotted line marks the estimated breakdown budget $\widehat{\delta}_{\mathsf{b}}$ for the sign of the effect, defined for $\widehat{\beta}_*$ as in (5.1). The bounds carry no sampling uncertainty (see Remark SA-18 in the supplementary materials).

already at $\delta = 0.25$ with them. Put differently, the gap between the OLS and the recentered IV estimates, which is what the recentering is designed to remove, can be reproduced by assignment processes that are not far from the postulated one. Third, the ST26 bounds are nested in ours even though they need not be in general. In this application, the nesting reflects the difference in the least (best) favorable assignment processes attaining each type of bound: the distributions attaining ours concentrate mass on a few configurations, which the multiplicative constraint in (6.2) rules out by capping the reweighting of every configuration at the factor ω.

7 Conclusion

This paper developed a framework for global sensitivity analysis in which the sensitivity parameter δ is the share of the population allowed to violate a distributional assumption. The sensitivity bounds are the values of finite-dimensional programs, and at $\delta = 1$ they coincide with the exact nonparametric sharp bounds. We proposed plug-in estimators of the bounds and established their consistency, limit distribution, and the validity of a bootstrap procedure. We also proposed summary measures that separate misspecification from sampling uncertainty and make studies comparable. Applications to attrition in randomized experiments, counterfactuals in structural models, and formula instruments illustrate the questions these tools answer: which of two experiments to trust, how much of a study's fragility is due to sampling uncertainty, and what a second round would have to spend to certify more robustness.

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